The bias interval of the elephant random walk

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Abstract

We prove the conjecture of Guérin, Laulin, Raschel and Simon that the bimodal initial biases of the superdiffusive elephant random walk form a possibly empty open interval centered at one half. Outside it, including its endpoints, the limit density has a unique local maximum. Finite sign bounds pass to the limit under weak convergence. They give exactly two inflection points for the fair unimodal density and four for the bimodal density. A moment argument proves bimodality for memory at least 0.79 without computation.

A certificate locates the unique fair-start transition near 0.66881286. The critical density has a quartic top. Its two modes separate on a square-root scale, while the bias interval opens on a three-halves scale. Above the threshold, the bias boundary is positive and continuously differentiable, with a variational formula and a unique optimizer. Thus strongly biased starts are unimodal at every fixed memory, although the boundary tends to zero as memory tends to one.

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1 Introduction

An elephant and an ant are playing hide and seek. The elephant counts while the ant hides among many identical anthills, then opens its eyes and knows at once which one the ant is in. The ant's shoes are outside the door.

(Anonymous)

Fix the first step of a superdiffusive elephant random walk. Guérin, Laulin, Raschel and Simon prove that its limit density is unimodal [8, Theorem 1.1]. Randomizing the first step mixes this density with its reflection. The mixture can have two peaks, although each component has one. They ask which biases give two, conjecturing an open interval

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centered at one half [8, Remark 2.1(b)]. We prove their conjecture and locate the unique memory threshold where two peaks first appear under a fair start. A related conjecture, eventual log-concavity, is disproved by Trinas and Valle in a recent preprint [10].

Guérin, Laulin, Raschel and Simon prove conditional unimodality by induction in the finite walk and then pass to its limit. We need an induction that allows two peaks and an arbitrary start. We carry the boundary histories in augmented rows, so the interior sign argument also controls signs entering at the ends. Restricting these rows to the half favored by the first step excludes a valley there. Lattice-window averages then transfer both shape bounds under weak convergence (Lemma A.7). For curvature, Gaussian smoothing reduces the problem to Descartes' rule for exponential sums. The rule is classical. The contribution is the boundary and dominant-half construction that makes it apply to the biased walk.

1.1 The walk and its parameters

The elephant walk [1] takes steps $X_1, \dots, X_N \in \{+1, -1\}$, with $+1$ meaning right. At each step it chooses uniformly from its earlier steps and repeats the chosen direction, except with probability ε, when it goes the other way. Write U_t for the step chosen at time $t+1$, and call it the parent of that step. We choose all parents and all reversal coins independently, also independently of X_1.

Put $A_N = \#\{i \le N : X_i = +1\}$ and $P_N(k) = \mathbb{P}(A_N = k)$. The position is $S_N = 2A_N - N$. We take a fair first step unless another bias is specified, so $P_N(k) = P_N(N-k)$. The notation $\mathbb{P}_+$ and $\mathbb{E}_+$ conditions on $X_1 = 1$. For even N, write $k_c = N/2$. All logarithms are natural.

For fixed memory put $a = 1 - 2\varepsilon \in (1/2, 1)$ and condition on $X_1 = 1$. The almost-sure limit of S_N/N^a has density h_a. With $\mathbb{P}(X_1 = 1) = q$, its density is

$$g_{a,q}(u) = q h_a(u) + (1-q) h_a(-u), \qquad g_a = g_{a,1/2}.$$

The copying probability in the original notation is $p = (1+a)/2$. In the finite recurrences we use $r_t(k) = \varepsilon + (1-2\varepsilon)k/t$.

1.2 Fixed memory

Theorem 1.1 (the bias interval, analytic)**.** *For every $a \in (1/2, 1)$ there is $b_a \in (0, 1/2]$ such that $g_{a,q}$ has exactly two local maxima if and only if $b_a < q < 1 - b_a$. Outside this interval, including its endpoints, it is unimodal with a unique local maximum. For a fair start the density has exactly two inflection points when unimodal and four when bimodal. For $a \ge 79/100$ it is bimodal. On the open set where $b_a < 1/2$, the boundary is continuously differentiable, and $b_a \to 0$ as $a \uparrow 1$.*

Theorems 3.1 and 3.6 prove the shape assertions, and Corollary 3.3 gives the moment criterion. The interior argument in Theorem 3.8 proves positivity and regularity where the interval is nonempty. Corollary 3.9 gives the limit at one. These arguments use no numerical certificate.

Theorem 1.2 (the transition, computer-assisted)**.** *There is a unique threshold a_* such that the bias interval is empty exactly when $a \le a_*$. Numerically, $a_* \approx 0.6688$. At a_* the fair density has a quartic top. Its two modes emerge at distance proportional to $\sqrt{a - a_*}$, and*

$$\tfrac{1}{2} - b_a \sim Q_*(a - a_*)^{3/2}, \qquad Q_* \approx 9.86698769.$$

The interval strictly widens immediately above a_*.

The certified enclosure is in Appendix B. The certificate establishes the signs and derivative bounds in Theorem 3.4 and Table 1. The quartic and three-halves laws then

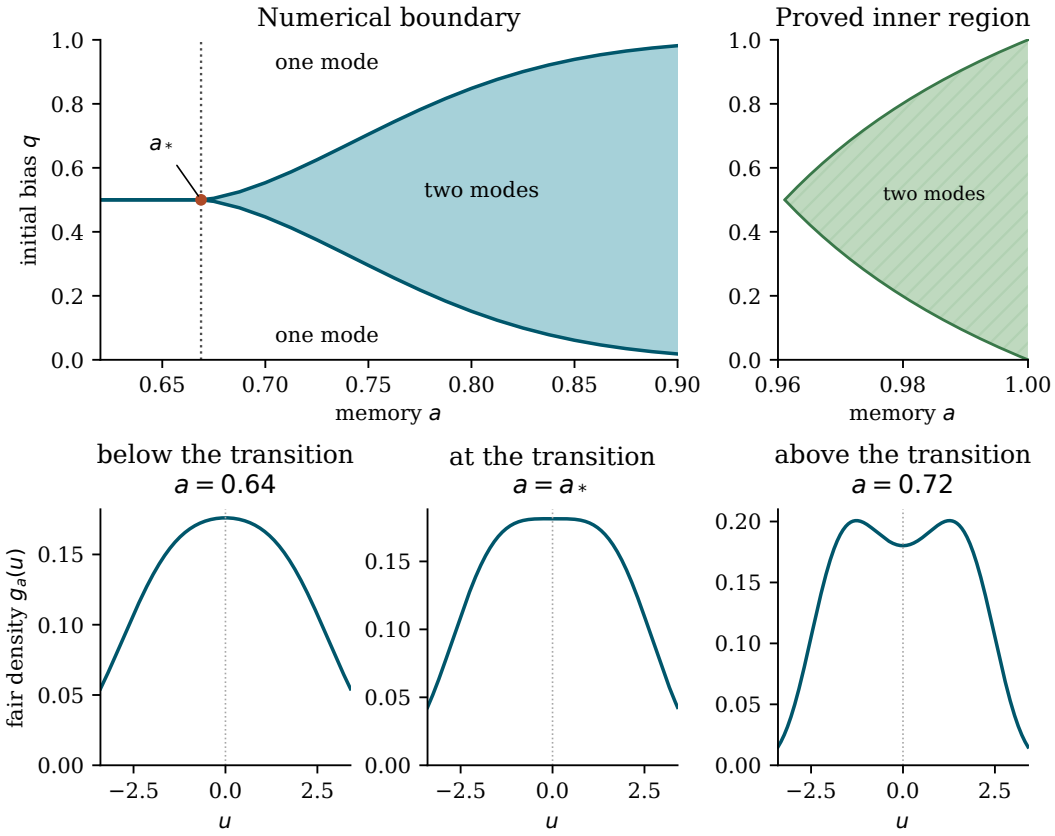


Figure 1: The bias interval and the fair-start transition. The upper left panel shows numerical values of the continuous boundary in Theorem 3.8 (the curves and shading are not certified enclosures). The upper right panel plots the explicit inner region $4\rho_a/(1-4\rho_a) < q < 1-4\rho_a/(1-4\rho_a)$ proved in Corollary 3.9. It does not locate the full boundary. The lower panels show the fair density before, at and after the transition. Its top is quartic at the threshold, whose certified enclosure is given in Appendix B.

follow by Taylor expansion (Propositions 3.10 and 3.11). Theorem 3.8 gives a continuous boundary, continuously differentiable and positive on $(a_*, 1)$, and a unique optimizer in its variational formula. Continuity at a_* uses the certified critical expansion. We do not prove that the boundary decreases throughout $(a_*, 1)$.

Theorem 1.3 (the finite shape bounds)**.** *For every $N \geq 1$, every $0 < \varepsilon < 1$ and every initial bias, the count law has at most two modal plateaus, including its endpoints. If $q \geq 1/2$, it dominates its reflection on the right half and has no strict valley there. For a fair start, $N \geq 3$ and $0 < \varepsilon < 1/2$, the nonzero signs of*

$$D_N(k) = P_N(k+1) - 2P_N(k) + P_N(k-1), \qquad 1 \leq k < N,$$

after equal neighbors are merged, form a subsequence of $(+,-,+,-,+)$.

The proofs are Proposition A.5, Proposition A.6 and Theorem 2.5. Weak convergence is enough to carry the valley bounds to a continuous limit (Lemma A.7). For the curvature bound, we first smooth by a Gaussian and then apply the sign bound.

1.3 Proof overview

Signs at the ends. The recurrence alone is not enough at the ends. We append the boundary end values to each row, so that the induction keeps track of where a new sign could enter. In this enlarged row, an entry is either copied or is a positive sum of neighbors from the preceding row. We can therefore bound the signs of the entire row. For first differences this gives at most two modal plateaus. For second differences under a fair start it gives at most five sign blocks.

The half favored by the first step. Take $q \geq 1/2$. The right half has no valley and dominates its reflection. A second peak therefore requires a decrease on the left, and this is also sufficient. Fix two points on the left and compare their values. The right value minus the left is affine in the bias and nonnegative at $q = 1$, by conditional unimodality. Once negative, it stays negative as the bias moves toward one half. Taking all such pairs gives an interval, and reflection gives its other half. Analyticity excludes flat pieces. We later use the two-mode bound again, this time to control the boundary ratio and prove continuity.

Passing to the limit. If the limit density had a forbidden valley, average over three short intervals around its high and low points. The inequalities remain strict, so weak convergence gives them for the finite law too. The smallest atom in the middle interval and the largest in each outer interval then give a forbidden finite valley. For two valleys we use five intervals instead (Lemma A.7). For curvature we first smooth the second differences with a Gaussian. Descartes' rule controls their signs. Summation by parts then gives the limit for a fixed Gaussian width, after which we let that width tend to zero.

Locating the threshold. We integrate the moment series for $g_a''(0)$ and bound its Fourier tail through the smoothing equation. Taylor models in memory retain the cancellations lost by direct interval evaluation. Near the zero, a positive parameter derivative makes the transition unique, while a negative fourth spatial derivative gives the quartic top. Table 1 records the certified constants. Appendices B and C prove every estimate used in the certificate.

Section 2 gives the finite bounds and Section 3 their limiting consequences. Sections 4 and 5 record verification scope and open questions. Longer proofs are in the appendices.

1.4 Prior work

The counting chain predates the elephant walk. In the formulation of James, James and Qi [4, p. 2339], the Drezner–Farnum model [2] has next-success probability $(1 - \theta)p + \theta k/t$. Taking $p = 1/2$ and $\theta = a$ gives our transition rule. Drezner and Farnum's family includes bimodal laws. Heyde [3] established the fixed-parameter limit regimes, and James, James and Qi treat time-varying dependence. We classify limiting shapes as the initial bias and memory vary.

Fixed-memory results include [5, 6, 7]. Murahara and Tokumitsu [11] give optimal Gaussian local and central limit rates for fixed copying probabilities at most $3/4$. Vishwakarma [12] studies moment recurrences in the three regimes. We use the density and moments of [9, Theorems 1.3 and 1.4] and conditional unimodality from [8, Theorem 1.1]. The local limit in Trinas and Valle's preprint [10, Theorem 3.1] could replace our weak-transfer argument. Their Theorem 4.4 disproves the log-concavity conjecture of [8, Remark 2.3].

The same transition appears in the growing-network model of Mohandas et al. [20]. They observe the symmetric bimodality transition and extrapolate its finite-length central

curvatures to a copying probability near 0.836 (Supplemental Material, Section IV.A). Our certified threshold is $(1 + a_*)/2 \approx 0.83440643$ in their parameter. The slow finite-length approach is visible in Remark 3.5.

The bias question also belongs to the classical study of modality in mixtures. Eisenberger [24] studies how the mixing weight affects bimodality for two normal components, and Robertson and Fryer [25] examine their shape across parameter space. Here the two components are a non-Gaussian limit density and its reflection. We prove that their bimodal weights form an interval by controlling the finite walk, without an explicit formula for the conditional density.

Dekking and Kong [18, Theorem 4.1] classify the Markov binomial law as unimodal, bimodal or trimodal. It depends on the previous state, so it differs from the walk here. Trinas [19] studies last returns to zero at fixed superdiffusive memory. His leading constant involves $g_{a,q}(0)$, which is independent of q by reflection. The bias changes the shape away from zero even though it leaves that value unchanged.

The sign arguments use classical variation diminution, as developed in Karlin's work [22, 21]. For Gaussian smoothing we use the generalized Descartes rule from Pólya and Szegő [23, Part V, Chapter 1]. We include the elementary forms needed here. The walk-specific step is to control the boundary histories, where the positive interior recurrence alone gives no sign bound.

2 Finite shape bounds

The second difference of P_N changes sign at most four times. Together with the first-difference bound, this controls the modes and inflection points of the limiting density. Appendix A treats first differences for arbitrary initial bias and proves the corresponding bounds for the limit density.

For the fair-start second differences, $0 < \varepsilon < \frac{1}{2}$ and $a = 1 - 2\varepsilon$, so that $r_t(b) = \varepsilon + ab/t$. For $t \geq 1$ and every integer k we write $P_t(k) = \mathbb{P}(A_t = k)$, which is 0 unless $0 \leq k \leq t$, and

$$D_t(k) = P_t(k+1) - 2P_t(k) + P_t(k-1).$$

We also put $s_t = D_t(1) = P_t(0) - 2P_t(1) + P_t(2)$ for $t \geq 2$.

Lemma 2.1 (recurrence for the second difference)**.** *For every $t \geq 1$ and every integer k,*

$$P_{t+1}(k) = \big(1 - r_t(k)\big)P_t(k) + r_t(k-1)\,P_t(k-1), \tag{2.1}$$

$$D_{t+1}(k) = \big(1 - r_t(k+1)\big)D_t(k) + r_t(k-2)\,D_t(k-1). \tag{2.2}$$

Also $P_t(k) = P_t(t-k)$ and $D_t(k) = D_t(t-k)$.

Proof. Given $A_t = k$, the copied step is $+1$ with probability k/t, so step $t+1$ is $+1$ with probability $(1-\varepsilon)\frac{k}{t} + \varepsilon(1-\frac{k}{t}) = r_t(k)$. This gives (2.1) for $0 \leq k \leq t+1$, where at $k = 0$ and at $k = t+1$ the formula has one extra term, namely $r_t(-1)P_t(-1)$ or $(1-r_t(t+1))P_t(t+1)$, and this term is 0 because $P_t(-1) = P_t(t+1) = 0$. For other k every term vanishes. The symmetry follows by reversing every step, which preserves the law under a fair start. Taking second differences gives the symmetry of D_t.

For (2.2) put $\varsigma = a/t$, so that $r_t(b+1) = r_t(b) + \varsigma$. We insert (2.1) at $k+1$, k and $k-1$ into the definition of $D_{t+1}(k)$ and collect the coefficients of $P_t(k+1)$, $P_t(k)$, $P_t(k-1)$ and $P_t(k-2)$. They are

$$1 - r_t(k+1), \qquad 3r_t(k) - 2, \qquad 1 - 3r_t(k-1), \qquad r_t(k-2).$$

The right side of (2.2) has the coefficients $1 - r_t(k+1)$, $-2 + 2r_t(k+1) + r_t(k-2)$, $1 - r_t(k+1) - 2r_t(k-2)$ and $r_t(k-2)$. The second ones agree because $2r_t(k+1) + r_t(k-2) = 3r_t(k)$, and the third ones agree because $r_t(k+1) + 2r_t(k-2) = 3r_t(k-1)$. □

Trinas and Valle [10, (2.2)] use (2.1), the forward equation, for a superdiffusive local limit theorem. Their higher-difference recurrence [10, (4.10)] supports a certified fixed-memory study of log-concavity. We use (2.2) to count sign changes. Its coefficients are positive for $2 \le k \le t-1$, so new sign changes can enter only at the ends. The next lemma is the elementary variation-diminishing fact behind this argument (see [22, 21] for the general theory).

For a finite real list z, define its sign pattern $\mathrm{sp}(z)$ by deleting the zeros of z, replacing each remaining entry by its sign and merging equal neighbors. Thus $\mathrm{sp}(z)$ lists the alternating signs. For two such lists we write $p \preceq q$ if p is a subsequence of q. We put $\Pi_5 = (+,-,+,-,+)$ and $\Pi_3 = (+,-,+)$.

Lemma 2.2 (positive sums of neighbors)**.** *Let $\zeta = (\zeta_1, \dots, \zeta_n)$ be real, and let $\eta_i = A_i\zeta_i + B_i\zeta_{i+1}$ for $1 \le i \le n-1$ with $A_i \ge 0$ and $B_i \ge 0$. Let $w = (\zeta_1, \eta_1, \zeta_2, \eta_2, \dots, \eta_{n-1}, \zeta_n)$. Then $\mathrm{sp}(w) = \mathrm{sp}(\zeta)$ and $\mathrm{sp}(\eta) \preceq \mathrm{sp}(\zeta)$.*

We prove this in Appendix D.1.

Proposition 2.3 (augmented rows and weak limits)**.** *Let laws on lattices of spacing $\eta_n \to 0$ converge weakly to a continuous density g. Include zero masses in each lattice row. Suppose that, for each n, its first differences occur in order in the last of a finite sequence of augmented rows. Each update has the form $z_i^{\mathrm{new}} = A_i z_i + B_i z_{i+1}$ with $A_i, B_i \ge 0$. If every initial row has sign pattern contained in $(+,-,+,-)$, then g has no two successive strict valleys. If the same construction for differences on the positive half-line starts with pattern contained in $(+,-)$, then g has no strict valley on $[0,\infty)$.*

Proof. Lemma 2.2 preserves the stated patterns through each finite sequence of rows. Two successive strict valleys would give differences with signs $-,+,-,+$, by telescoping. A single valley gives $-,+$. The respective patterns exclude these. Lemma A.7 transfers the exclusions to g by averaging over short lattice windows. □

For the walk, the work is to construct these rows and check their initial signs. Figure 2 shows the construction for first differences. Proposition A.5 checks the full rows, and Proposition A.6 checks the half favored by the first step.

$$\underbrace{\ell_N, \dots, \ell_{t+1}}_{\text{left boundary history}} \qquad \underbrace{d_t(0), \dots, d_t(t-1)}_{\text{active row}} \qquad \underbrace{u_{t+1}, \dots, u_N}_{\text{right boundary history}}$$

$$\downarrow \text{ copy or sum adjacent entries}$$

$$\ell_N, \dots, \ell_{t+2} \qquad \ell_{t+1}, d_{t+1}(1), \dots, d_{t+1}(t-1), u_{t+1} \qquad u_{t+2}, \dots, u_N$$

Figure 2: Fix a final length N and retain the boundary values that will enter later. Here $d_t(k) = P_t(k+1) - P_t(k)$, $\ell_t = d_t(0)$ and $u_t = d_t(t-1)$. The active row grows by one while the augmented row shrinks by one. The incoming ends are copied from the histories, so they cannot introduce an untracked sign change.

Lemma 2.4 (sign changes next to an end)**.** *For every $t_1 \ge 2$ we have $\mathrm{sp}(s_2, s_3, \dots, s_{t_1}) \preceq \Pi_3$. Also, for every $0 \le \varepsilon \le 1$, the numbers s_3 and s_4 do not both vanish.*

We prove this in Appendix D.2.

Theorem 2.5 (at most five blocks)**.** *Let $N \ge 3$ and $0 < \varepsilon < \frac{1}{2}$. Then*

$$\mathrm{sp}\big(D_N(1), D_N(2), \dots, D_N(N-1)\big) \preceq (+,-,+,-,+).$$

So the second difference of P_N changes sign at most four times on $\{1, \dots, N-1\}$.

We prove this in Appendix D.3.

3 The limit density at fixed memory

Fix $\varepsilon \in (0, \frac{1}{4})$, so $a = 1 - 2\varepsilon \in (\frac{1}{2}, 1)$. Conditional on $X_1 = 1$, S_k/k^a converges almost surely to $\mathcal{L}_a$ [6, Theorem 3.7]. The fair-start limit $\mathcal{L}_a^\circ$ has the law of $\xi\mathcal{L}_a$, where ξ is a fair sign independent of $\mathcal{L}_a$ [9, (1.3) and (1.6)]. Guérin, Laulin and Raschel prove that both limits have positive, bounded and smooth densities [9, Theorem 1.3], denoted by h_a and g_a as before. Guérin, Laulin, Raschel and Simon prove that $\mathcal{L}_a$ is unimodal [8, Theorem 1.1].

3.1 The number of modes

A probability density h on $\mathbb{R}$ is *unimodal* if for some m it is nondecreasing on $(-\infty, m]$ and nonincreasing on $[m, \infty)$. Guérin, Laulin, Raschel and Simon ask for a unique local maximum [8], which also rules out a flat top. The density g_a is a mixture of the unimodal density of $\mathcal{L}_a$ and its reflection, but mixing unimodal densities can give more than one peak [15]. First take a fair start. Lemma A.4 rules out constant intervals, so the modes are points. We consider other biases afterward.

Theorem 3.1 (at most two modes)**.** *Let $a \in (\frac{1}{2}, 1)$. Exactly one of the following holds.*

(a) The density g_a is nondecreasing on $(-\infty, 0]$ and nonincreasing on $[0, \infty)$.

(b) There is $\kappa > 0$ such that g_a is nondecreasing on $[0, \kappa]$ and nonincreasing on $[\kappa, \infty)$, and $g_a(0) < g_a(\kappa)$.

In case (b), g_a is nondecreasing on $(-\infty, -\kappa]$ and nonincreasing on $[-\kappa, 0]$, and it is not unimodal. All these monotonicities are strict. The unique local maximum is 0 in case (a), and the only local maxima are $\pm\kappa$ in case (b). In case (a), there is $d > 0$ such that g_a is strictly concave on $(-d, d)$ and strictly convex on each tail. In case (b), there are $0 < d_1 < \kappa < d_2$ such that g_a is strictly convex on $(-d_1, d_1)$ and on each tail beyond $\pm d_2$, and strictly concave on $(-d_2, -d_1)$ and (d_1, d_2). Thus it has exactly two inflection points in case (a) and four in case (b).

We cannot pass directly from the second differences of P_N to the curvature of g_a. First smooth with a Gaussian and apply Descartes' rule to the resulting exponential sum. Now let $N \to \infty$, keeping the width fixed, and only then send the width to 0. Appendix A proves the lemmas used below. A real function v has *values at most* Π if $\operatorname{sp}(v(y_1), \dots, v(y_j)) \preceq \Pi$ for all $y_1 < \cdots < y_j$, and *increments at most* Π if $\operatorname{sp}(v(y_2) - v(y_1), \dots, v(y_j) - v(y_{j-1})) \preceq \Pi$ for all such points. We put $\Pi_4 = (+, -, +, -)$.

Proof. Fix a, so that $\varepsilon = (1 - a)/2 \in (0, \frac{1}{4})$.

Step 1, smoothing. Fix $\sigma > 0$ and let $\varphi(y) = e^{-y^2/(2\sigma^2)}/(\sigma\sqrt{2\pi})$. For $N \geq 3$ put $\eta = 2N^{-a}$ and $z_k = (2k - N)N^{-a}$, so that $S_N/N^a = z_{A_N}$, and let

$$V_N(y) = \eta^{-2} \sum_{k=1}^{N-1} D_N(k)\, \varphi(y - z_k).$$

Since $\varphi(y - z_k) = \varphi(y) e^{-z_k^2/(2\sigma^2)} e^{y z_k/\sigma^2}$ and the z_k increase with k, Lemma A.1 and Theorem 2.5 show that V_N has values at most Π_5. (If every $D_N(k)$ vanishes, then $V_N = 0$ and every pattern is empty.)

Step 2, the limit $N \to \infty$. Let $u(y) = \mathbb{E}\varphi(y - \mathcal{L}_a^\circ)$. The derivatives of φ are bounded, so u is smooth and $u^{(j)}(y) = \mathbb{E}\varphi^{(j)}(y - \mathcal{L}_a^\circ)$. We show that $V_N(y) \to u''(y)$ for every y. Put $P_N(k) = 0$ for $k \notin \{0, \dots, N\}$. Then $D_N(k) = 0$ unless $-1 \leq k \leq N + 1$, and summation by parts gives

$$\sum_{k=-1}^{N+1} D_N(k)\, \varphi(y - z_k) = \sum_{k=0}^{N} P_N(k)\, \Delta\varphi(y - z_k), \qquad \Delta\varphi(t) = \varphi(t + \eta) - 2\varphi(t) + \varphi(t - \eta).$$

Taylor's formula gives $|\eta^{-2}\Delta\varphi(t) - \varphi''(t)| \le \eta^2\|\varphi''''\|_\infty/12$. By symmetry the four terms with $k \in \{-1, 0, N, N+1\}$ on the left are at most $\varphi(0)\,(6P_N(0) + 2P_N(1))$ in absolute value. Here $P_N(0) = \frac{1}{2}(1-\varepsilon)^{N-1}$. The boundary recurrence gives $\xi_{n+1} = (1 - a/((1-\varepsilon)n))\xi_n + \varepsilon/(1-\varepsilon)$, where $\xi_n = P_n(1)/P_n(0)$ and $\xi_1 = 1$. Thus $\xi_n \le 1 + (n-1)\varepsilon/(1-\varepsilon)$ and $P_N(1) = O_\varepsilon(N(1-\varepsilon)^N)$. As $\eta^{-2} = N^{2a}/4$, these four terms times η^{-2} tend to 0. Hence

$$V_N(y) = \mathbb{E}\varphi''\big(y - S_N/N^a\big) + o(1) \longrightarrow \mathbb{E}\varphi''(y - \mathcal{L}_a^\circ) = u''(y),$$

because $S_N/N^a \to \mathcal{L}_a^\circ$ in law and φ'' is bounded and continuous. By Lemma A.2, u'' has values at most Π_5.

Step 3, integrating the sign bound. By dominated convergence, $u'(y) \to 0$ as $y \to \pm\infty$. So u has increments at most Π_4, by Lemma A.3.

Step 4, the limit $\sigma \to 0$. Write $u = u_\sigma$. As g_a is bounded and continuous, $u_\sigma(y) = \int g_a(y-t)\varphi(t)\,dt$ tends to $g_a(y)$ for every y. So the increments of u_σ tend to those of g_a, and Lemma A.2 shows that g_a has increments at most Π_4. Lemma A.4 also makes g_a'' bounded and continuous, so $u_\sigma'' = g_a'' * \varphi \to g_a''$ pointwise. Hence g_a'' has values at most Π_5.

Step 5, symmetry. Let $0 \le y_1 < \cdots < y_j$, let $d_i = g_a(y_{i+1}) - g_a(y_i)$, and let p be the pattern of the d_i. As g_a is even, the points $-y_j < \cdots < -y_1 < y_1 < \cdots < y_j$, with $-y_1$ left out if $y_1 = 0$, have the increments $-d_{j-1}, \ldots, -d_1$, then 0 if $y_1 > 0$, then $d_1, \ldots, d_{j-1}$. Their pattern is the reverse of $-p$ followed by p. It has $2|p|$ entries and is $\preceq \Pi_4$. So p is empty, $(+)$, $(-)$ or $(+,-)$. In particular there are no $0 \le y < y' < y''$ with $g_a(y) > g_a(y') < g_a(y'')$.

If every increment on $[0,\infty)$ is nonpositive, case (a) holds. Otherwise choose $y_1 < y_2$ with $g_a(y_1) < g_a(y_2)$. Integrability supplies $y_3 < y_4$, both beyond y_2, with $g_a(y_3) > g_a(y_4)$. The sign bound excludes any later increase, so g_a attains a maximum on $[0,\infty)$ at some κ. The absence of a strict valley makes it nondecreasing before κ and nonincreasing afterward. The positive increment forces $\kappa > 0$. If $g_a(0) = g_a(\kappa)$, it would be constant on $[0,\kappa]$, contrary to analyticity and the positive increment. Thus $\kappa > 0$ and case (b) holds. Evenness gives the left-half shape and excludes unimodality in (b).

Every stated monotonicity is strict. Otherwise equal values at two points would force a constant interval, excluded by Lemma A.4. For the same reason, there are no additional local maxima.

Step 6, curvature. The Fourier bound in Lemma A.4 gives $g_a(y), g_a'(y) \to 0$ as $y \to \infty$. Evenness and the Π_5 bound give at most three sign blocks for g_a'' on $(0,\infty)$. Its last sign must be positive. Otherwise g_a' would eventually be nonincreasing with limit zero, making g_a eventually nondecreasing. This contradicts $g_a > 0$ and $g_a \to 0$. Analyticity and strict monotonicity imply that the first nonconstant term in the even Taylor series of g_a at zero is negative in case (a) and positive in case (b). The same signs therefore hold for g_a'' near zero. In case (b), g_a'' must also be negative somewhere, since g_a' is positive near zero and negative somewhere beyond κ. The positive-half-line sign patterns are consequently $(-,+)$ in case (a) and $(+,-,+)$ in case (b). Analyticity excludes an interval of zero curvature and makes the resulting convexities strict. In case (b), $g_a'(d_1) > 0 > g_a'(d_2)$, so $d_1 < \kappa < d_2$. □

Theorem 3.2 (two modes for memory near one)**.** *If $\frac{4}{5} \le a < 1$, that is $0 < \varepsilon \le \frac{1}{10}$, then case (b) of Theorem 3.1 holds. So g_a has exactly two modes.*

Proof. In case (a), Khinchine's representation [15] gives $\mathcal{L}_a^\circ \overset{d}{=} UZ$, with U uniform on $(0,1)$ and independent of Z. Hence Cauchy–Schwarz gives $\mathbb{E}(\mathcal{L}_a^\circ)^4 \ge \frac{9}{5}[\mathbb{E}(\mathcal{L}_a^\circ)^2]^2$. The moment recursion of [9, Theorem 1.4] instead gives

$$\frac{\mathbb{E}\mathcal{L}_a^4}{(\mathbb{E}\mathcal{L}_a^2)^2} = \frac{6(2a^2+2a-1)}{4a-1}\,\mathrm{Beta}(2a,2a),$$

where Beta is the beta function. For $a \geq \frac{3}{4}$, its integral and $t(1-t) \leq \frac{1}{4}$ give $\mathrm{Beta}(2a, 2a) \leq 4^{-2a+3/2}\,\mathrm{Beta}(\frac{3}{2}, \frac{3}{2}) = \pi 2^{-4a}$. Put $R(a) = (2a^2 + 2a - 1)/(4a - 1)$. On $[\frac{3}{4}, 1]$ we have $(\log R)' \leq 2 < 4 \log 2$. The first inequality reduces to $4a(4a^2 + a - 2) \geq 0$. Thus $6\pi R(a) 2^{-4a}$ decreases there, and for $a \geq \frac{4}{5}$ the displayed moment ratio is at most

$$\frac{282\pi}{55\, 2^{16/5}} < \frac{141}{80} < \frac{9}{5},$$

using $\pi < 22/7$ and $2^{1/5} > 8/7$. Even moments are unchanged by symmetrization, so this contradicts case (a). Theorem 3.1 then gives case (b). □

Corollary 3.3 (a larger computation-free range)**.** *The conclusion of Theorem 3.2 holds for* $79/100 \leq a < 1$.

Proof. The preceding upper bound $6\pi R(a) 2^{-4a}$ decreases for $a \geq 3/4$. At $a = 79/100$, use $2^{4/25} > 111/100$, which follows from the exact integer inequality $111^{25} < 16 \cdot 100^{25}$. Substitute this bound and $\pi < 22/7$ to get

$$6\pi R(79/100) 2^{-79/25} < \frac{33517}{18648} < \frac{9}{5}.$$

So the kurtosis is again too small for a unimodal density, proving the claim. □

We locate the transition through the curvature at the origin. The shape theorem and certified signs leave only a short interval to check. There the central curvature strictly increases with memory, so it has at most one zero.

Theorem 3.4 (the threshold, computer-assisted)**.** *Put*

$$z_- = 1838418790980431/2748779069440000 = 0.668812860014597754\ldots,$$
$$z_+ = 45960469774511/68719476736000 = 0.668812860014601028\ldots,$$

so that $z_+ - z_- < 3.3 \cdot 10^{-15}$.

(a) There is exactly one $a_* \in (\frac{1}{2}, 0.6712]$ *with* $g''_{a_*}(0) = 0$, *and* $z_- < a_* < z_+$.

(b) For $\frac{1}{2} < a \leq a_*$, *case (a) of Theorem 3.1 holds. For* $a_* < a < 1$, *case (b) holds. Thus* a_* *is the unique transition from one mode to two, and the critical density is unimodal.*

(c) We have $g''_a(0) < 0$ *for* $\frac{1}{2} < a < a_*$ *and* $g''_a(0) > 0$ *for* $a_* < a \leq 0.6712$.

Thus the transition is unique even inside the bracket. For $[79/100, 1)$ we also have the moment proof of Corollary 3.3, which needs no computation. For a biased start we lose symmetry, and instead use first differences.

Remark 3.5 (the certificate and finite-length thresholds)**.** Appendix B bounds the Fourier integral for $g''_a(0)$ by Taylor models in a and a tail estimate from the fixed-point equation. It uses scaled moments near $a = \frac{1}{2}$ and a Hankel test for $a \geq 0.6705$. The certificate takes about 2.8 hours of CPU time. On the exact finite law, $D_N(N/2)$ changes sign at $a \approx 0.6563$ for $N = 4000$ and at $a \approx 0.6592$ for $N = 8000$. These two numbers are computed and are not used in any proof. They illustrate how slowly the finite law approaches the limiting transition.

3.2 Arbitrary initial bias

Write h_a for the density of $\mathcal{L}_a$ and $g_{a,q}(x) = qh_a(x) + (1-q)h_a(-x)$ for initial bias $\mathbb{P}(X_1 = 1) = q$. Thus $g_{a,1/2} = g_a$.

Theorem 3.6 (the interval of bimodal biases)**.** *For every $a \in (1/2, 1)$ and $q \in [0,1]$, the density $g_{a,q}$ has at most two local maxima. There is $b_a \in [0, 1/2]$ such that it has exactly two local maxima if and only if*

$$b_a < q < 1 - b_a.$$

Outside this open interval, including its endpoints, the density is unimodal with a unique local maximum. The interval is empty exactly when $a \le a_*$, in which case $b_a = 1/2$. At finite time, for every $0 < \varepsilon < 1$, $q \in [0,1]$ and $N \ge 1$, the count law has at most two modal plateaus, including its endpoints.

Proof. The mode bound. Let P_N here denote the count law with first-step bias q. The finite assertion follows from Proposition A.5. In particular, the count law cannot have two successive strict valleys. Neither can $g_{a,q}$. If it did, weak convergence of S_N/N^a and averages over five short intervals would give five finite-law atoms with those same forbidden inequalities (Lemma A.7). There are also no constant intervals, for any bias, by Lemma A.4. Hence three local maxima $x < y < z$ would give $x < u < y < v < z$ with

$$g_{a,q}(u) < \min\{g_{a,q}(x), g_{a,q}(y)\}, \qquad g_{a,q}(v) < \min\{g_{a,q}(y), g_{a,q}(z)\}.$$

For each pair, choose a point in a one-sided maximizing neighborhood of the smaller maximum. There is a point with strictly smaller value, since the density is not constant on an interval. The resulting two valleys contradict the limit bound.

The bias interval. Fix a, write $f_q = g_{a,q}$ and $h = h_a$, and call $x < y < z$ a strict valley if $f_q(y) < \min\{f_q(x), f_q(z)\}$. Proposition A.6 and the same weak-transfer lemma show that, for $q \ge 1/2$, f_q has no strict valley on $[0, \infty)$ and $f_q(-x) \le f_q(x)$ for $x \ge 0$. Consequently it has a strict valley if and only if $f_q(x) > f_q(y)$ for some $x < y \le 0$. Such a descent gives the valley $(x, y, -x)$ by reflection dominance. Conversely, a valley must start at $x < 0$. If its middle point y is positive, the no-valley property at $(0, y, z)$ gives $f_q(0) \le f_q(y) < f_q(x)$.

At $q = 1$, conditional unimodality [8] excludes a valley, so h is nondecreasing on $(-\infty, 0]$. For fixed $x < y \le 0$, the difference $f_q(x) - f_q(y)$ is affine in q and nonpositive at $q = 1$. If positive at $q \ge 1/2$, it remains positive for every $r \in [1/2, q]$. The valley biases in $[1/2, 1]$ are therefore downward closed. They are relatively open, since the strict inequalities at three fixed points persist under a small change of bias, and they exclude 1. Reflection gives an open interval $(b_a, 1 - b_a)$, possibly empty.

A strict valley gives two local maxima. Integrability and positivity supply an outer point on each side below the corresponding witness height, and we maximize on the two resulting compact intervals. Conversely, two local maxima give a strict valley because there is no flat interval. The mode bound excludes a third maximum. A continuous integrable density without a strict valley is unimodal, and absence of flat intervals makes its maximum unique. Finally the bias interval is nonempty exactly when it contains $1/2$, so Theorem 3.4 identifies the empty case. □

This proves the initial-bias conjecture of [8, Remark 2.1(b)], with its half-width $q(a) = 1/2 - b_a$. Trinas and Valle's stronger local limit [10, Theorem 3.1], in an unrefereed arXiv preprint, would also transfer the finite inequalities. Weak convergence suffices for the proof here.

3.3 The bias boundary

To find the endpoint of the bias interval, we compare the slopes of the conditional density at reflected points. This reduces the problem to a one-dimensional infimum.

Proposition 3.7 (a variational equation for the bias boundary)**.** *For $a \in (1/2, 1)$ put $U_a = \{t > 0 : h'_a(t) > 0\}$. With the infimum over an empty set interpreted as $+\infty$,*

$$b_a = \min\left\{\frac{1}{2}, \quad \inf_{t \in U_a} \frac{h'_a(-t)}{h'_a(t) + h'_a(-t)}\right\}. \tag{3.1}$$

In particular $b_a \to 0$ as $a \uparrow 1$. If the infimum is attained at $t > 0$ with $h'_a(t) > 0$ and $b_a < 1/2$, then, with $q = 1 - b_a$,

$$g'_{a,q}(-t) = g''_{a,q}(-t) = 0.$$

At this point the smaller peak disappears.

Proof. The proof of Theorem 3.6 shows that $h'_a(-t) \geq 0$ for $t \geq 0$. For $q \geq 1/2$ bimodality is equivalent to a decrease somewhere on the negative half-line, hence to $g'_{a,q}(-t) < 0$ for some $t > 0$. Here

$$g'_{a,q}(-t) = q h'_a(-t) - (1 - q) h'_a(t).$$

This expression can be negative only if $t \in U_a$, in which case it is negative exactly when $1 - q > h'_a(-t)/(h'_a(t) + h'_a(-t))$. Taking the union over t gives (3.1), with equality excluded as in the interval theorem. At an interior minimizer the ratio has derivative zero. Differentiating its numerator and denominator gives the two displayed equations. The moment estimate in Corollary 3.9 gives $b_a \to 0$. □

Theorem 3.8 (regularity of the bias boundary)**.** *The function $a \mapsto b_a$ is continuous on $(1/2, 1)$, is continuously differentiable on $(a_*, 1)$, and extends continuously to $b_1 = 0$. For every $a > a_*$, we have $0 < b_a < 1/2$, and the infimum in (3.1) has a unique minimizer $t_a \in (0, \kappa_a)$. This minimizer depends continuously on a throughout $(a_*, 1)$.*

Proof. For $a > a_*$, the curvature partition in Theorem 3.1 gives $g'_a(t) > 0$ on $(0, \kappa_a)$ and $g'_a(t) < 0$ on (κ_a, ∞). For $t \geq \kappa_a$ in U_a, the ratio in (3.1) is therefore at least $1/2$. Since $b_a < 1/2$, only $(0, \kappa_a)$ can lower the infimum. This restricts the search before we assert that a minimizer exists. Put

$$D_a(t) = h'_a(t) + h'_a(-t), \qquad R_a(t) = \frac{g'_a(t)}{D_a(t)} \quad (0 < t < \kappa_a).$$

Lemmas A.8 and A.9 give $h'_a(-t) > 0$ for every $t \geq 0$. We therefore have $D_a > 0$, $0 < R_a < 1/2$, and $1/2 - b_a = \sup_{0<t<\kappa_a} R_a(t)$. Also $R_a(0+) = 0$.

The ratio cannot have a strict valley. More precisely,

$$R_a(v) \geq \min\{R_a(u), R_a(w)\} \qquad (0 < u < v < w < \kappa_a). \tag{3.2}$$

Otherwise, choose $s \in (0, 1/2)$ strictly between the middle value and both outer values. With $q = 1/2 + s$, $g'_{a,q}(-t) = D_a(t)(s - R_a(t))$ then has signs $-, +, -$ at $-w, -v, -u$, and is positive sufficiently close to zero on the left. Positivity of the density and its zero limits at infinity supply a positive derivative farther left and a negative derivative farther right. There would be three local maxima, contrary to Theorem 3.6.

At κ_a, we have $D_a(\kappa_a) = 2h'_a(-\kappa_a) > 0$. Thus R_a extends continuously to $[0, \kappa_a]$, with value zero at both endpoints. Its positive maximum is attained in the interior. At that point $h'_a(-t) > 0$, so the maximum is strictly below $1/2$. This is the compactness step needed for $b_a > 0$, since pointwise strict inequality on an open interval would not suffice. Equation (3.2) and analyticity make this maximum unique, since two maximizers would force a constant interval. They also make R_a strictly increasing before its maximizer and strictly decreasing after it.

Lemma A.4 gives joint continuity of all the derivatives used here. The signs of g'_a on either side of its unique positive zero give continuity of κ_a. Fix $a_0 > a_*$ and choose $0 < l < t_{a_0} < r < \kappa_{a_0}$. The strict inequalities $R_a(l), R_a(r) < R_a(t_{a_0})$ persist near a_0. For $v < l$, apply (3.2) to v, l, t_{a_0}. Since $R_a(l) < R_a(t_{a_0})$, it gives $R_a(v) \le R_a(l)$. Applying it to t_{a_0}, r, v gives $R_a(v) \le R_a(r)$ for $v > r$. Thus every nearby global maximum lies in the fixed interval $[l, r]$. Uniform convergence on this fixed compact interval proves continuity of the maximum and, by uniqueness, of its location. This proves continuity above a_*. Below it $b_a = 1/2$, Proposition 3.11 gives continuity at a_*, and Proposition 3.7 gives the limit at one.

For differentiability, the same three-point argument puts all nearby maximizers in one compact interval. If $M(a) = 1/2 - b_a$, maximality gives, for $s > 0$,

$$R_{a+s}(t_a) - R_a(t_a) \le M(a+s) - M(a) \le R_{a+s}(t_{a+s}) - R_a(t_{a+s}).$$

Divide by s and use the continuity of t_a and $\partial_a R_a$ from Lemma A.4. The argument for $s < 0$ is the same. Thus

$$-b'_a = \frac{\partial_a h'_a(t_a)\, h'_a(-t_a) - h'_a(t_a)\, \partial_a h'_a(-t_a)}{\{h'_a(t_a) + h'_a(-t_a)\}^2}. \tag{3.3}$$

The right side is continuous. No derivative of t_a is needed. □

We can use this argument to enclose the boundary numerically, since three comparisons restrict the search to a bounded interval. Verified inequalities $R_a(l), R_a(r) < R_a(t)$ with $0 < l < t < r < \kappa_a$ trap the maximum in $[l, r]$. If $L \le R_a(t)$ and $R_a \le U$ throughout that interval, then $1/2 - U \le b_a \le 1/2 - L$. If these inequalities hold uniformly, the enclosure covers a whole parameter interval. More concretely, if $|R_a(s) - R_{a_0}(s)| \le M|a - a_0|$ for $s = l, t, r$, the two comparisons at a_0 hold throughout $|a - a_0| \le w$ whenever their gaps exceed $2Mw$. To prove strict decrease of b_a on this interval, it would suffice to have

$$R_{a_2}(s) - R_{a_1}(s) \ge c(a_2 - a_1) \quad (l \le s \le r,\ a_1 < a_2), \qquad c > 0.$$

Indeed, evaluate the later ratio at the earlier maximizer. This is an interval criterion, not a verified sign bound for the full curve.

Corollary 3.9 (a rate for the bias boundary near one)**.** *Put*

$$\rho_a = \frac{\Gamma(1+a)^2}{(2a-1)\Gamma(2a)} - 1.$$

If $\rho_a < 1/12$*, then*

$$0 \le b_a \le \frac{4\rho_a}{1 - 4\rho_a}.$$

In particular, as $a \uparrow 1$*,* $b_a \le 8(1-a) + O((1-a)^2)$*.*

Proof. The conditional limit has mean $\mu_a = 1/\Gamma(1+a) > 0$ and variance $\rho_a \mu_a^2$, by the moment formulas of [9, Theorem 1.4]. Consider the three adjacent intervals of length μ_a centered at $-\mu_a$, 0 and μ_a. Chebyshev's inequality puts at least $1 - 4\rho_a$ of the conditional law in the right interval and at most $4\rho_a$ in the middle. For a start to the left, reflect these intervals. After mixing, the left interval has mass at least $(1-q)(1-4\rho_a)$ and the right at least $q(1-4\rho_a)$. The middle still has mass at most $4\rho_a$.

For $4\rho_a/(1-4\rho_a) < q < 1 - 4\rho_a/(1-4\rho_a)$, the middle has less mass than either outer interval. All three intervals have the same length. So, by continuity, we can choose a point in each with the middle value strictly smaller than the other two. This rules out unimodality. The two-mode bound in Theorem 3.6 gives exactly two modes, and therefore the stated bound on b_a. Finally, the logarithmic derivative at $a = 1$ of $1 + \rho_a$ is $2\psi(2) - 2 - 2\psi(2) = -2$. Thus $\rho_a = 2(1-a) + O((1-a)^2)$, which gives the linear bound. □

3.4 The density at the threshold

We next ask how far apart the modes are just after the transition. The needed constants are derivatives at the threshold. Write

$$C_* = \partial_a g_a''(0)|_{a=a_*}, \qquad B_* = -g_{a_*}^{(4)}(0), \qquad H_* = h'_{a_*}(0).$$

Table 1 gives strict certified enclosures. The coefficient Q_* is defined in Proposition 3.11.

Constant	Lower bound	Upper bound
C_*	0.6889503945	0.6889503947
B_*	0.079207784468	0.079207784470
H_*	0.09707494510086	0.09707494510088
Q_*	9.86698768	9.86698770
$\sqrt{6C_*/B_*}$	7.224131934	7.224131936

Table 1: The constants in the critical expansions. The enclosure is uniform on a parameter interval containing the certified bracket for a_* (Appendix C.4).

Proposition 3.10 (the critical shape)**.** *The density at the threshold satisfies*

$$g_{a_*}(u) = g_{a_*}(0) - \frac{B_*}{24}u^4 + O(u^6) \qquad (u \to 0).$$

For $a > a_*$ let $\kappa_a > 0$ be the positive mode of g_a. As $a \downarrow a_*$,

$$\kappa_a = \sqrt{\frac{6C_*}{B_*}}\sqrt{a - a_*} + o\big(\sqrt{a - a_*}\big),$$
$$g_a(\kappa_a) - g_a(0) = \frac{3C_*^2}{2B_*}(a - a_*)^2 + o\big((a - a_*)^2\big).$$

Proof. On a neighborhood of a_*, the certificate bounds $|\phi_a(t)|$ by Ct^{-8} and $|\partial_a\phi_a(t)|$ by $4t^{-7}$ for large t (Appendix C.4). On a bounded frequency interval we instead use the moment series, which converges locally uniformly, as does its parameter derivative. Together these bounds let us pass limits and derivatives through the Fourier integral. We get spatial derivatives through order six continuous in a and one parameter derivative of the central curvature. On a smaller parameter interval, the Taylor remainders through order four are uniform. At criticality, evenness, $g''_{a_*}(0) = 0$ and $g_{a_*}^{(4)}(0) = -B_* < 0$ give the first assertion.

Put $\delta = a - a_* > 0$. Uniform Taylor expansion gives

$$\frac{g'_a(u)}{u} = C_*\delta + o(\delta) - \frac{B_* + o(1)}{6}u^2 + O(u^4).$$

For $u = k\sqrt{\delta}$ this has the sign of $C_* - B_*k^2/6$ when k is fixed away from $\sqrt{6C_*/B_*}$. The unique positive mode lies between any two such choices with opposite signs, by Theorem 3.1. Squeezing gives the mode expansion. Substitute it into

$$g_a(u) - g_a(0) = \tfrac{1}{2}\big(C_*\delta + o(\delta)\big)u^2 - \tfrac{1}{24}\big(B_* + o(1)\big)u^4 + O(u^6)$$

to obtain the height expansion. □

Proposition 3.11 (how the bias interval opens)**.** *With the constants of Table 1, put*

$$Q_* = \frac{\sqrt{2}\,C_*^{3/2}}{3H_*\sqrt{B_*}}, \qquad Q_* \approx 9.86698769.$$

As $a \downarrow a_*$,

$$\frac{1}{2} - b_a = Q_*(a - a_*)^{3/2} + o\big((a - a_*)^{3/2}\big). \tag{3.4}$$

On some interval immediately above a_*, b_a is strictly decreasing. For $a > a_*$ sufficiently close to a_*, the infimum in (3.1) is attained. The minimizer t_a of Theorem 3.8 satisfies

$$t_a = \sqrt{\frac{2C_*}{B_*}}\sqrt{a - a_*} + o\big(\sqrt{a - a_*}\big) = \frac{\kappa_a}{\sqrt{3}} + o\big(\sqrt{a - a_*}\big).$$

At $q = 1 - b_a$, the stationary point is $-t_a$.

Proof. Proposition 3.10 supplies the required regularity. The refined certificate in Appendix C.4 encloses $\pi^{-1}\int_0^\infty t \operatorname{Im}\phi_a(t)\,dt = h_a'(0)$ uniformly on an interval containing a_*, giving the positive H_* in Table 1.

Write $\delta = a - a_*$. By (3.1),

$$\frac{1}{2} - b_a = \max\left\{0, \sup_{t \in U_a} \frac{g_a'(t)}{h_a'(t) + h_a'(-t)}\right\}.$$

The fair-shape theorem makes the numerator nonpositive for $t \geq \kappa_a$. Proposition 3.10 gives $\kappa_a = O(\sqrt{\delta})$, and $H_* > 0$ makes both conditional derivatives positive on this shrinking interval. Uniform Taylor expansion gives, for bounded $v \geq 0$,

$$\frac{g_a'(v\sqrt{\delta})}{h_a'(v\sqrt{\delta}) + h_a'(-v\sqrt{\delta})} = \frac{\delta^{3/2}}{2H_*}\left(C_* v - \frac{B_* v^3}{6}\right) + o(\delta^{3/2}).$$

The cubic is largest at the unique positive point $v = \sqrt{2C_*/B_*}$. Maximizing on $[0, \kappa_a/\sqrt{\delta}]$, whose right endpoint tends to $\sqrt{6C_*/B_*}$, proves (3.4). The maximum is positive and attained inside $(0, \kappa_a)$, since the ratio vanishes at both endpoints. Uniform convergence and the unique maximum of the limiting cubic also force every scaled maximizer to tend to $\sqrt{2C_*/B_*}$. This proves the assertion about t_a. The bounds on Q_* follow by squaring its formula and inserting the three rational enclosures for C_*, B_* and H_*.

For strict monotonicity, choose a small fixed $r > 0$. For a sufficiently close to a_*, the denominator of the ratio is positive on $[0, r]$ and $\kappa_a < r$. The maximum can therefore be taken over this fixed compact interval. The differentiated Fourier bounds justify the expansions $\partial_a g_a'(t) = C_* t + o(t)$ and $\partial_a(h_a'(t) + h_a'(-t)) = O(1)$, uniformly as $(a, t) \to (a_*, 0)$, while $g_a'(t) = O((|a - a_*| + t^2)t)$. The parameter derivative of the ratio is consequently positive for $0 < t \leq r$ after shrinking the parameter interval and r. At $a > a_*$ its maximum is positive and occurs at $t > 0$. Evaluating at that maximizer at any larger nearby memory makes the maximum strictly larger. Hence b_a strictly decreases there. □

This also explains why two different powers appear. Near zero, the derivative is approximately $C_*(a - a_*)t - B_* t^3/6$. Its positive part has width of order $\sqrt{a - a_*}$ but height of order $(a - a_*)^{3/2}$. The modes separate on the width scale. The bias must overcome the height to remove the smaller peak.

4 Verification

Table 2 separates analytic arguments from interval computations. The numerical illustrations are not used in any proof.

Claim	How it is established	Trusted base
Finite shape bounds	Written recurrence and sign proofs	Exact finite identities and ordered sign patterns
Fair-limit shape, inflection counts, bias interval and boundary regularity	Written shape, transfer and maximization proofs	Cited limit law, conditional unimodality and Fourier representation
Unique memory threshold	Interval certificate and written Fourier estimates	Arb transcendental arithmetic and the stated uniform envelopes
Quartic critical top, emerging modes and bias width	Written Taylor and maximization arguments using certified derivatives	The same Fourier envelopes and interval enclosures

Table 2: Proofs and their inputs. We do not use the numerical illustrations in the proofs. Appendix F gives the reproduction records.

The shared formalization uses Lean 4.33.1 and Mathlib at `0df444a360ea`. Its audit checks 386 modules and $24{,}005$ declarations, allowing only `propext`, `Classical.choice` and `Quot.sound`. The accompanying MAP.md records each statement, its hypotheses and whether it belongs to this paper or the archived combined version. Lean checks finite recurrences, reflection comparisons and several deductions from named analytic hypotheses. It does not replay the transcendental interval enclosures. Boundary regularity and the critical expansions have written proofs. Thus the audit checks the formal statements listed in the map, not every argument in the manuscript.

The transition check retains its acceptance tests under `python -O`. A second check evaluates the two threshold endpoint signs directly at the real parameter values. It bypasses the parameter Taylor models, though it still uses Arb and the same analytic tail bound.

5 Open problems

The full bias boundary. Is b_a strictly decreasing on $(a_*, 1)$? It decreases near a_* (Proposition 3.11), and Figure 1 suggests the same elsewhere. Formula (3.3) reduces this to a sign question at the unique optimizer, but we do not know its sign globally. The moment estimate gives $b_a \le 8(1-a) + O((1-a)^2)$. What is the exact decay near one?

A finite-length threshold. Does the central curvature have exactly one zero in $(0, 1)$ at every even length? The computations in Remark 3.5 suggest a threshold increasing toward a_*. The question at every length is separate from the limiting bias interval proved here.

The Drezner–Farnum model and higher dimensions. Drezner and Farnum's family includes bimodal laws [2]. Changing its innovation probability p changes the transition rule, while changing the first-step bias q mixes two reflected laws. Our mixture argument therefore does not settle the shapes for $p \neq 1/2$. In higher dimensions, the fixed-point theory of [9] gives densities, but our sign argument needs points ordered along a line. Even the meaning of a central depression needs care (the usual axis-step walk is not rotationally symmetric). We have not computed either extension.

Acknowledgments. I thank Peter Kagey, whose Problem 131 of the Open Problem Collection [13] led to the crossing question, and Aidan Deshong for helpful conversations. The opening riddle recalls the crossing question that led to this work, namely to look for the first reversal.

A note on the use of AI

I wrote the original draft of this paper and its mathematics. I used Claude Opus 5.5 (Anthropic, through Claude Code) for literature searches, citation checks and referee-style reviews that informed the framing and abstract. I read every source cited in the paper. I proposed and drafted the following additions: Lemma A.4; the strictness and inflection-point clauses of Theorem 3.1; the fourth-moment extension of Theorem 3.2 to $a \geq 4/5$; the Jensen lower bound in Theorem 4.5(a) of the archived combined version; and the weaker weak-convergence hypothesis in its Theorem 5.9(c). I used GPT-6 Astra (OpenAI, through Codex) to revise the manuscript and to revise the proofs of these additions. Proofs aside from the aforementioned were not altered by either model. The associated Lean files in kagey-problems/problem131/lean/PaperD were originally written by me and edited using GPT-6 Astra through Codex. Section 4 distinguishes the formal proofs, their named classical hypotheses and the separate numerical checks. I take responsibility for the claims, proofs and computations in the paper.

Code and data

The tagged repository contains the source, certificate records and Lean files. The submission package also contains the numerical records and checksums. The archived combined version retains the rare-reversal results. For the assumptions and formal coverage, see Section 4.

A Sign patterns and regularity

We use the notation of Sections 2 and 3.1 to prove the lemmas for Theorem 3.1. Deleting entries can only remove signs from a list's pattern. An alternating list p satisfies $p \preceq q$ for every alternating list q with the same first sign and at least as many entries.

The following is the exponential-sum form of the generalized Descartes rule [23, Part V, Chapter 1]. We give its short proof to specify the sign-pattern conclusion used below.

Lemma A.1 (Descartes' rule for exponential sums)**.** *Let $\lambda_1 < \cdots < \lambda_n$ and $b_1, \ldots, b_n$ be real, with $b = (b_1, \ldots, b_n) \neq 0$, and put $F(y) = \sum_{k=1}^{n} b_k e^{\lambda_k y}$. Then F has finitely many real zeros, and $\operatorname{sp}(F(y_1), \ldots, F(y_j)) \preceq \operatorname{sp}(b)$ for all $y_1 < \cdots < y_j$. The same holds for ψF for every positive function ψ.*

Proof. Let $Z(F)$ be the number of distinct real zeros of F, and $S(b)$ the number of sign changes of b. We show $Z(F) \leq S(b)$ by induction on $S(b)$. If $S(b) = 0$, then F has the sign of the nonzero b_k everywhere. If $S(b) \geq 1$, choose i such that the last nonzero b_k with $k \leq i$ and the first nonzero b_k with $k > i$ have opposite signs, and choose $\lambda_i < \kappa < \lambda_{i+1}$. Then

$$G(y) = \frac{d}{dy}\big(e^{-\kappa y} F(y)\big) = \sum_{k=1}^{n} b_k (\lambda_k - \kappa) e^{(\lambda_k - \kappa) y}.$$

The coefficient $b_k(\lambda_k - \kappa)$ has the sign of $-b_k$ for $k \leq i$ and of b_k for $k > i$, so the coefficients of G have $S(b) - 1$ sign changes. By Rolle's theorem G vanishes between any two zeros of F, so $Z(F) \leq Z(G) + 1 \leq S(b)$.

If b_{k_0} is the first nonzero b_k, then $e^{-\lambda_{k_0} y} F(y) \to b_{k_0}$ as $y \to -\infty$. The zeros of F cut the line into at most $S(b) + 1$ open intervals, on each of which F has one sign. With equal neighbors merged, these signs form an alternating list that starts with the sign of b_{k_0} and has at most $S(b) + 1$ entries, so it is $\preceq \operatorname{sp}(b)$. The nonzero values $F(y_1), \ldots, F(y_j)$ lie in these intervals in order. A positive factor ψ changes no sign. □

Lemma A.2 (pointwise limits)**.** *Let v_n and v be real functions on $\mathbb{R}$ with $v_n(y) \to v(y)$ for every y, and let Π be an alternating list. If every v_n has values at most Π, then so has v. If every v_n has increments at most Π, then so has v.*

Proof. Let $y_1 < \cdots < y_j$, and let I be the set of indices i with $v(y_i) \neq 0$. For all large n, $v_n(y_i)$ has the sign of $v(y_i)$ for every $i \in I$. Then $\operatorname{sp}(v(y_1), \ldots, v(y_j))$ is the pattern of the list $(v_n(y_i))_{i \in I}$, which is a subsequence of $\operatorname{sp}(v_n(y_1), \ldots, v_n(y_j)) \preceq \Pi$. The increments of v_n at $y_1, \ldots, y_j$ tend to those of v, and the same argument applies to them. □

Lemma A.3 (integrating the sign bound)**.** *Let u be twice continuously differentiable on $\mathbb{R}$, with $u'(y) \to 0$ as $y \to \pm\infty$. If u'' has values at most Π_5, then u' has values at most Π_4, and u has increments at most Π_4.*

Proof. It is enough to treat points $y_1 < \cdots < y_j$ at which the values of u' are nonzero and alternate in sign. Let s be the sign of $u'(y_1)$. Choose $y_0 < y_1$ with $|u'(y_0)| < |u'(y_1)|$, and $y_{j+1} > y_j$ with $|u'(y_{j+1})| < |u'(y_j)|$. For $0 \le i \le j$ the mean value theorem gives $w_i \in (y_i, y_{i+1})$ with $u''(w_i)(y_{i+1} - y_i) = u'(y_{i+1}) - u'(y_i)$. The right side has the sign s for $i = 0$, the sign of $u'(y_{i+1})$ for $1 \le i \le j-1$, and the sign of $-u'(y_j)$ for $i = j$. In each case this is $(-1)^i s$. So u'' takes $j+1$ alternating signs, starting with s, at $w_0 < \cdots < w_j$. This list is $\preceq \Pi_5$, so $j + 1 \le 5$, and $j + 1 \le 4$ if s is $-$. In both cases the list of the j signs of $u'(y_1), \ldots, u'(y_j)$ is $\preceq \Pi_4$.

For the increments at $y_1 < \cdots < y_j$, each nonzero $u(y_{i+1}) - u(y_i)$ has the sign of $u'(\xi_i)$ for some $\xi_i \in (y_i, y_{i+1})$, by the mean value theorem. So the pattern of the increments is a pattern of values of u'. □

Lemma A.4 (entire densities and parameter continuity)**.** *For every $a \in (\frac{1}{2}, 1)$ and $1 < r < 1/a$, the characteristic function ϕ_a of $\mathcal{L}_a$ satisfies*

$$|\phi_a(t)| \le 2e^{-c|t|^r} \qquad (t \in \mathbb{R})$$

for some $c = c(a, r) > 0$. The same c works on a sufficiently small neighborhood of any fixed a. Every spatial derivative of h_a and its first derivative in a are jointly continuous in (a, u). For every $q \in [0, 1]$, the density of the limit with first-step bias q extends to an entire function and is not constant on an interval of positive length.

Proof. First, we check weak continuity in a. On a compact $J \subset (1/2, 1)$, the martingale $\mathcal{M}_n(a) = \Gamma(n)S_n/\Gamma(n + a)$ of [6, (2.4), Appendix A] satisfies

$$\sup_{a \in J} \mathbb{E}|\mathcal{L}_a - \mathcal{M}_n(a)|^2 \le C_J \sum_{k \ge n} k^{-2 \min J} \longrightarrow 0.$$

The walk's centered next step has conditional variance at most one, and the gamma ratios are uniformly bounded on J. The finite laws are continuous in a, so $\phi_a(t)$ is jointly continuous, uniformly on compact frequency intervals.

The fixed-point equation of [9, Theorem 1.1 and (2.10)] gives

$$|\phi_a(t)| \le \int_0^1 |\phi_a(tv^a)|\,|\phi_a(t(1-v)^a)|\,dv.$$

Here ϕ_a is the characteristic function conditional on a positive first step. The random sign disappears after taking absolute values, since $|\phi_a(-s)| = |\phi_a(s)|$. Fix a_0 and $1 < r < 1/a_0$, and choose J about a_0 with $r \max J < 1$. Put $D(v) = v^{r \max J} + (1-v)^{r \max J} - 1$. We have $D(v) > 0$ for $0 < v < 1$. Dominated convergence therefore permits $K > 1$ such that

$$R := 2 \int_0^1 e^{-K^r (\log 2) D(v)}\,dv < 1.$$

The law of $\mathcal{L}_{a_0}$ has a density [9, Theorem 1.3], so the Riemann–Lebesgue lemma gives $T > 0$ with $|\phi_{a_0}(t)| < 2^{-K^r}$ for $|t| \geq T$. Joint continuity, after shrinking J, gives $|\phi_a(t)| \leq 2^{1-K^r}$ for $a \in J$ and $T \leq |t| \leq KT$. Set $c = (\log 2)/T^r$ and $b(t) = 2e^{-ct^r}$ for $t \geq 0$. On $[0, T]$ we have $b \geq 1$, and on $[T, KT]$ we have $b \geq 2^{1-K^r}$. Thus $|\phi_a| \leq b$ on $[0, KT]$.

For $t \geq KT$,

$$\frac{\int_0^1 b(tv^a)b(t(1-v)^a)\,dv}{b(t)} \leq 2\int_0^1 e^{-ct^r D(v)}\,dv \leq R < 1.$$

Suppose the interval on which $|\phi_a| \leq b$ holds has finite supremum $t_* \geq KT$. By continuity it holds on $[0, t_*]$ as well. At the endpoint the fixed-point inequality gives the strict bound $|\phi_a(t_*)| \leq Rb(t_*) < b(t_*)$. Continuity then extends the interval past t_*, a contradiction. The bound thus holds on $[0, \infty)$, and evenness extends it to $\mathbb{R}$.

Since $r > 1$, the Fourier integral converges absolutely and locally uniformly for complex arguments. Indeed, for every $B < \infty$, $\int_{\mathbb{R}} e^{B|t|}|\phi_a(t)|\,dt < \infty$. The Fourier integral therefore defines an entire function. After any fixed number of spatial derivatives, the bound on J still supplies an integrable majorant, giving joint continuity by dominated convergence. For initial bias q, apply the triangle inequality to $q\phi_a(t) + (1-q)\phi_a(-t)$ to get the same bound. A density of integral one cannot be constant on $\mathbb{R}$. If it were constant on an interval, the identity theorem would make it constant everywhere.

For the memory derivative, Lemma C.1(b) gives a complex moment majorant and local parameter differentiation on every compact frequency interval. The differentiated smoothing inequalities in Lemma C.9 extend this to Fourier inversion. Indeed, for any prescribed power β, the exponential bound above supplies $f(t) \leq Ct^{-\beta}$. Taking T large makes both contraction coefficients R and κ in that lemma smaller than one. Choose D_1 to bound the frequency derivative on $[T_2, T]$ and to satisfy its strict contraction inequality, then choose D_a in the same way for the memory derivative. The maximum argument in that proof gives $|\partial_a \phi_a(t)| = O_J((1+|t|)^{1-\beta})$. Here β is arbitrary and the compact memory interval can be shrunk about any fixed parameter. Multiplication by any fixed power of t therefore leaves an integrable majorant. Differentiation under Fourier inversion proves the asserted joint continuity. □

A.1 First differences for a biased start

Proposition A.5 (two modes for every initial bias)**.** *Let $0 < \varepsilon < 1$, let $q \in [0,1]$ be the probability that the first step is positive, and let $P_n(k)$ be the law of the number of positive steps after $n \geq 1$ steps. Extend P_n by zero outside $\{0, \dots, n\}$. The first differences $d_n(k) = P_n(k+1) - P_n(k)$ satisfy*

$$\operatorname{sp}\big(d_n(-1), d_n(0), \dots, d_n(n)\big) \preceq (+,-,+,-).$$

In particular, P_n has at most two modal plateaus, including endpoint modes. Moreover, there are no integers $i_0 < \cdots < i_4$ for which

$$P_n(i_0) > P_n(i_1) < P_n(i_2) > P_n(i_3) < P_n(i_4).$$

Proof. Put $a = 1 - 2\varepsilon$ and $r_t(k) = \varepsilon + ak/t$. The forward equation is

$$P_{t+1}(k) = (1 - r_t(k))P_t(k) + r_t(k-1)P_t(k-1), \qquad P_1 = (1-q, q).$$

Subtracting adjacent equations gives

$$d_{t+1}(k) = r_t(k-1)d_t(k-1) + (1 - r_t(k+1))d_t(k), \qquad 1 \leq k \leq t-1. \tag{A.1}$$

Both coefficients are at least $\min(\varepsilon, 1-\varepsilon) > 0$, since $r_t(j)$ is a convex combination of ε and $1-\varepsilon$ for $0 \leq j \leq t$. It remains to check the two ends.

Suppose first that $q < 1$. Since $P_t(0) = (1-q)(1-\varepsilon)^{t-1} > 0$, define

$$\xi_t = \frac{P_t(1)}{P_t(0)}, \qquad b = \frac{1-2\varepsilon}{1-\varepsilon} < 1, \qquad A = \frac{\varepsilon}{1-\varepsilon} > 0.$$

The boundary equation gives

$$\xi_{t+1} = (1 - b/t)\xi_t + A, \qquad \xi_1 = \frac{q}{1-q}.$$

Thus $\delta_t = \xi_{t+1} - \xi_t$ obeys

$$\delta_{t+1} = \frac{t-b}{t+1}\delta_t + \frac{A}{t+1}. \tag{A.2}$$

Once an increment is nonnegative, every later increment is positive. Thus ξ_t first decreases and then increases (either part may be empty). For $\ell_t = d_t(0) = P_t(0)(\xi_t - 1)$ this proves

$$\operatorname{sp}(\ell_1, \dots, \ell_N) \preceq (+, -, +).$$

If $q \le 1/2$, then $\xi_1 \le 1$ and the sharper bound is $\operatorname{sp}(\ell_1, \dots, \ell_N) \preceq (-, +)$. In particular, a positive ℓ_i cannot be followed by a negative ℓ_j. Zeros do not affect either conclusion.

Write $u_t = d_t(t-1)$. Reflection of the walk gives $u_t(q) = -\ell_t(1-q)$ and $d_t^{(1-q)}(t-1-k) = -d_t^{(q)}(k)$. For $0 < q \le 1/2$, the right boundary history therefore has pattern contained in $(-, +, -)$. For $q = 0$ it is nonpositive throughout. The conditional count is at most $t-1$, so $u_t = -P_t(t-1) \le 0$.

We now keep the boundary values that the later rows will need. Fix N and, for $1 \le t \le N$, form the augmented row

$$Z_t = (\ell_N, \dots, \ell_{t+1}, d_t(0), \dots, d_t(t-1), u_{t+1}, \dots, u_N).$$

The left history is read backward and the right history forward. Thus an endpoint entering the active row is already present in the preceding augmented row. This is why an interior recurrence suffices once the two histories have been checked. For $q \le 1/2$, its first row is contained in $(+, -, +, -)$. To include possible zeros at $\ell_1 = u_1 = 2q - 1$, argue as follows. If Z_1 contained four ordered signs $-, +, -, +$ and the second lay in the reversed left history, the first two would give a positive left edge followed in time by a negative one, which is impossible. Otherwise the last three lie in the right history and have signs $+, -, +$. For $q > 0$, their reflected negatives give the forbidden pattern $(-, +, -)$ in the left history at bias $1 - q$. For $q = 0$, they contradict the nonpositive right history. This also covers $N = 1$.

Every entry of Z_{t+1} is a nonnegative combination of adjacent entries of Z_t. The interior entries use (A.1). The two new boundary entries and the remaining future edge values are copied, with coefficients $(1, 0)$ or $(0, 1)$. To check the ordering, number the augmented entries from left to right. Output entry i uses input entries i and $i+1$, including copied entries by allowing one coefficient to vanish. A nonzero output has a parent of the same sign. For consecutive opposite output signs, the chosen parent indices cannot coincide, since one input cannot have both signs. They are therefore strictly increasing. Applying this to any alternating subsequence of the output embeds its signs, in order, in the input. No alternating pattern can be added. Thus $\operatorname{sp}(Z_N) \preceq (+, -, +, -)$. Reflection and sign reversal preserve this pattern, proving the claim also for $q \ge 1/2$, including $q = 1$.

Finally, extension by zero adds $d_N(-1) = P_N(0) \ge 0$ on the left and $d_N(N) = -P_N(N) \le 0$ on the right. Neither enlarges the pattern. Every modal plateau contributes a positive-to-negative change, so there are at most two. If the last displayed inequalities in the statement held, each of their four intervals would contain an adjacent

difference of the same strict sign, by telescoping. These four differences would have signs $-,+,-,+$, a contradiction. □

Proposition A.6 (the dominant half)**.** *Under the hypotheses of Proposition A.5, suppose $q \geq 1/2$. Then $P_n(n-k) \leq P_n(k)$ for $2k \geq n$, and the right half has no strict valley. That is,*

$$P_n(j) \geq \min\{P_n(i), P_n(k)\} \qquad \text{if } i < j < k \text{ and } 2i+1 \geq n.$$

Equivalently, its first differences contain no negative value followed by a positive one. Reflection gives the corresponding left-half result when $q \leq 1/2$.

Proof. Write the bias explicitly as $P_{n,q}$ and put

$$A_n(k) = P_{n,q}(k) - P_{n,1-q}(k) = P_{n,q}(k) - P_{n,q}(n-k).$$

The forward equation gives $A_{n+1}(k) = (1 - r_n(k))A_n(k) + r_n(k-1)A_n(k-1)$. At $n = 1$, $A_1(1) = 2q - 1 \geq 0$. Suppose the assertion holds at length n. At a point strictly above the new center, both parents lie in the preceding upper half and both coefficients are nonnegative. At an integer center $A_{n+1}(k) = 0$ by reflection. Terms outside the support vanish. This proves the induction. In particular, the central first difference of an odd row is nonnegative.

Restrict the augmented row Z_t in the preceding proof to spatial indices $2k + 1 \geq t$. At $t = 1$ its pattern is contained in $(+,-)$. The central entry is $2q - 1 \geq 0$, followed by the future right boundary values $u_s(q) = -\ell_s(1-q)$. Since $1 - q \leq 1/2$, their pattern is $(+,-)$ by (A.2). We exclude a negative difference followed by a positive one, including the future boundary entries. If such a pair first appeared in the next row, each entry would have a same-sign parent. Their indices must be strictly ordered by the preceding proof. The sole possible parent outside the previous half occurs at the new central difference of an odd row. That entry is nonnegative, so it cannot begin the forbidden pair. More explicitly, if a negative entry at index i of row $t+1$ satisfies $2i + 1 \geq t + 1$, then $2i > t$, so both its parents satisfy $2k + 1 \geq t$. Parents of a later positive entry follow them strictly, as in the preceding proof. Induction therefore excludes the pair in every restricted augmented row. A strict valley would contain such two adjacent differences by telescoping, which proves the claim. □

Lemma A.7 (weak transfer of shape)**.** *Let probability laws on lattices $s_n + \eta_n\mathbb{Z}$, with $\eta_n > 0$ and $\eta_n \to 0$, converge weakly to a law with continuous density g. If their lattice mass sequences, including zero masses, exclude two successive strict valleys, then so does g. If they exclude a strict valley among positive lattice points, then g has no strict valley on $[0,\infty)$. For reflection-symmetric lattices, if the mass at each positive lattice point is at least that at its reflection, then $g(x) \geq g(-x)$ for $x \geq 0$.*

Proof. For a bounded interval J of positive length, let $K_n(J)$ be its lattice indices and let $\mu_n(J)$ be its probability. Lattice counting gives $|\eta_n \# K_n(J) - |J|| \leq 2\eta_n$, so these sets are eventually nonempty. Since the limiting law has no atoms, weak convergence gives

$$B_n(J) := \frac{\mu_n(J)}{\eta_n \# K_n(J)} \longrightarrow \frac{1}{|J|}\int_J g(x)\,dx.$$

Here $B_n(J)$ averages the scaled atom masses on J. If g had two successive strict valleys, choose five disjoint equal-length intervals about the witnesses. Choose their common radius so that every value on a high interval exceeds every value on either neighboring low interval. This is possible because there are only four strict comparisons and g is continuous. Their averages therefore retain all four inequalities. The corresponding B_n

satisfy these inequalities for large n. Choose a point of maximum mass in the first, third and fifth intervals, and one of minimum mass in each of the other two. The intervals are disjoint and ordered, so the selected points are ordered too. Each selected high mass is at least its interval average, and each selected low mass is at most its interval average. Thus the same four strict comparisons hold between individual atom masses. Including zero masses is essential here, since a low interval need not meet the support of a particular finite law. This is a contradiction.

Three positive intervals give the one-valley assertion by the same maximum–minimum–maximum selection. If the first witness is 0, continuity moves it slightly to the right while preserving the strict inequalities. Finally, for any interval $J \subset (0, \infty)$ the reflection hypothesis gives $\mu_n(-J) \leq \mu_n(J)$. Weak convergence passes this inequality to the limiting interval masses. Shrink J to $x > 0$ and use continuity to get $g(-x) \leq g(x)$. At $x = 0$ there is equality. □

A.2 A start with one step in each direction

Lemma A.8. *Fix the first two steps to be $+1, -1$. For $0 \leq a \leq 1$, the count law at each length $n \geq 2$ is log-concave, with positive support $\{1, \dots, n-1\}$. If $1/2 < a < 1$, its limiting density f_a^{+-} is symmetric and log-concave, with $(f_a^{+-})'(t) < 0$ for $t > 0$. Consequently the density conditional on a positive first step satisfies*

$$h_a'(t) > 0 \qquad (t < 0).$$

Proof. The extra initial step removes the boundary obstruction to the log-concavity induction of [8, Proposition 2.2]. To see this directly, write one update as $g_k = Af_k + Bf_{k-1}$, where

$$A = \frac{1+a}{2} - \frac{ak}{n}, \qquad B = \frac{1-a}{2} + \frac{a(k-1)}{n}, \qquad d = \frac{a}{n}.$$

For $u = f_{k-2}$, $v = f_{k-1}$, $w = f_k$ and $z = f_{k+1}$, expansion gives

$$\begin{aligned} g_k^2 - g_{k-1}g_{k+1} = (A^2 - d^2)(w^2 - vz) + (B^2 - d^2)(v^2 - uw) \\ + (A - d)(B - d)(wv - uz) + d^2(w - v)^2. \end{aligned}$$

For $2 \leq k \leq n-1$ we have $A, B \geq d \geq 0$. Log-concavity and interval support make all three gaps nonnegative. At either boundary, an adjacent output mass vanishes. The initial row is a point mass at one, so induction proves the assertion. The same recurrence gives strict positivity throughout the support.

Log-linear interpolation of these lattice masses is log-concave. Its integral differs from that of the step interpolation by at most twice the largest atom, which tends to zero because the limiting law has a continuous density. Thus the limit is log-concave. The two-subtree representation gives

$$\mathcal{L}_a^{+-} \overset{d}{=} U^a \mathcal{L}_a' - (1-U)^a \mathcal{L}_a'',$$

with independent copies and an independent uniform U. This proves symmetry. It also gives positivity and an entire density, by positivity of h_a and the Fourier bound of Lemma A.4. The derivative of $\log f_a^{+-}$ is nonincreasing and vanishes at zero. If it vanished at a positive point, the density would be constant on an interval, contradicting analyticity. Hence $(f_a^{+-})' < 0$ on the positive half-line.

For two positive initial steps, reflection dominance and the first-difference recurrence give nonnegative first differences on the negative half-line. The induction starts from the point mass at count two. Both parents stay in the preceding lower half except at

the central difference of an odd row, where reflection dominance gives the required sign. Passing to the derivative limit as in Lemma E.1 gives $(f_a^{++})'(t) \geq 0$ for $t < 0$. Finally $h_a = ((1+a)/2)f_a^{++} + ((1-a)/2)f_a^{+-}$. The second summand has strictly positive derivative there. □

Lemma A.9 (the conditional mode is positive)**.** *For every $1/2 < a < 1$, we have $h_a'(0) > 0$. In particular the unique conditional mode lies strictly to the right of zero.*

Proof. We keep a positive amount of the finite reflection difference when passing to the limit. Write $f_n(k) = \mathbb{P}_+(A_n = k)$ and, on $z \equiv n \pmod 2$, put $d_n(z) = f_n((n+z)/2) - f_n((n-z)/2)$. The reflection induction in Proposition A.6 is strict for $0 < z \leq n$ when the start is positive. Every noncentral upper-half entry has a positive parent with positive coefficient, starting from $d_1(1) = 1$. The recurrence is

$$d_{n+1}(z) = A_n(z)d_n(z+1) + B_n(z)d_n(z-1),$$
$$A_n(z) = \frac{1}{2} - \frac{a(z+1)}{2n}, \qquad B_n(z) = \frac{1}{2} + \frac{a(z-1)}{2n}.$$

Fix $n_0 > 4a$ and divide by z. After multiplication by $\prod_{j=n_0}^{n-1}(1-2a/j)^{-1}$, the resulting values $v_n(z)$ satisfy a backward averaging equation with weights

$$q_+ = \frac{A_n(z)(z+1)}{z(1-2a/n)}, \qquad q_- = \frac{B_n(z)(z-1)}{z(1-2a/n)}.$$

The weights sum to one and are nonnegative in the interior band used below. At $z = 1$, the weight of the parent zero vanishes. For the corresponding backward chain,

$$\mathbb{E}[z_n^2 \mid z_{n+1} = z] = \frac{n-4a}{n-2a}z^2 + \frac{3n-4a}{n-2a}.$$

Set $\gamma_n = \Gamma(n-2a)/\Gamma(n-4a) \sim n^{2a}$. Then $Y_n = z_n^2/\gamma_n$ has backward drift $\delta_n = (3n-4a)/\{(n-2a)\gamma_n\}$, and $\sum_{n\geq n_0}\delta_n \to 0$ as $n_0 \to \infty$.

Fix $R > 0$. Start at $z_N = 1$ for odd N and $z_N = 2$ for even N, and stop at row n_0 or on leaving $0 < z/\sqrt{\gamma_n} \leq R$. For sufficiently large n_0, every band state and its immediate parents lie inside the support. The expected stopped value of Y is at most $z_N^2/\gamma_N + \sum_{n\geq n_0}\delta_n$. At an exit it exceeds R^2, so increasing n_0 and then N makes the exit probability less than one half. All stopped v values are positive. Their minimum m_0 at row n_0 is positive, since that row has finitely many band states. The bounded stopping argument gives $v_N(z_N) \geq m_0/2$. The product normalizer is asymptotic to a positive constant times N^{2a}, and hence

$$\liminf_N N^{2a}\frac{d_N(z_N)}{z_N} > 0.$$

The first-difference convergence proved in Lemma E.1, applied to the conditional start, identifies the limit as $4h_a'(0)$. Conditional unimodality then puts its mode to the right of zero. □

B Certifying the memory threshold

This appendix describes the certificate for Theorem 3.4. Appendix C proves the estimates it uses. Fix $a \in (\frac{1}{2}, 1)$ and write $\phi(t) = \mathbb{E}e^{it\mathcal{L}_a}$. Put $\theta_j = 1$ for even j and $\theta_j = a$ for odd j, let $m_0 = m_1 = 1$ and $m_n = (na - \theta_n)^{-1}\sum_{j=1}^{n-1}\theta_j m_j m_{n-j}$ for $n \geq 2$. Then $\mathbb{E}\mathcal{L}_a^n = n!\,c_n$ with $c_n = m_n/\Gamma(1+an)$ [9, Theorem 1.4]. As $\mathbb{E}e^{t|\mathcal{L}_a|} < \infty$ for real t [9, Corollary 2.11], $\phi(t) = \sum_n c_n(it)^n$ for all real t. By [9, Theorem 1.1 and (2.10)], $\mathcal{L}_a$ has

the law of $V^a\mathcal{L}'_a + \xi(1-V)^a\mathcal{L}''_a$, with V uniform on $[0,1]$, $\xi = \pm 1$, $\mathcal{L}'_a$ and $\mathcal{L}''_a$ distributed as $\mathcal{L}_a$, and all four independent. Since $|\phi(-s)| = |\phi(s)|$, this gives

$$|\phi(t)| \le \int_0^1 |\phi(tv^a)|\,|\phi(t(1-v)^a)|\,dv \qquad (t \in \mathbb{R}). \tag{B.1}$$

The density g_a of $\mathcal{L}_a^\circ$ has the characteristic function $\operatorname{Re}\phi$.

Step 1, one value of a. Lemmas C.1 and C.2 bound the moment-series remainder by a geometric series. Lemma C.3 then bounds $|\phi|$ on frequency cells. For $T_3 \ge 2^a T_2$, $\beta > 3$ and $a\beta > 1$, Lemma C.4 gives an explicit test $R < 1$ that propagates $|\phi(t)| \le Ct^{-\beta}$ from $[T_2, T_3]$ to $[T_2, \infty)$. We integrate the moment series for $g''_a(0) = -\pi^{-1}\int_0^\infty t^2 \operatorname{Re}\phi(t)\,dt$ on $[0, T_3]$ and bound both the series remainder and Fourier tail by Lemma C.5.

Step 2, intervals of a. Direct interval evaluation loses cancellations between terms of size about 10^{60}. Instead, on $I = [a_0 - w, a_0 + w]$ we expand in $a - a_0$. Choose $\rho = w/\vartheta$ with $0 < \vartheta < 1$. A complex bound $|h| \le M$ on the disc of radius ρ gives the degree-K remainder $M\vartheta^{K+1}/(1-\vartheta)$ on I. The complex moment bound and gamma product formula in Proposition C.7 give a polynomial G and η with $|g''_a(0) - G(a - a_0)| \le \eta$ throughout I.

Step 3, a near $\frac{1}{2}$. Set $y = 2a - 1$, $\mu_n = m_n y^{\lfloor n/2 \rfloor}$ and $\sigma^2 = \mathbb{E}\mathcal{L}_a^2 = 1/(y\Gamma(2a))$. The scaled moments are analytic for $\operatorname{Re} a > 1/4$ (Lemma C.8). Applying Steps 1 and 2 to $\mathcal{L}_a/\sigma$ bounds $\sigma^3 g''_a(0)$ up to $a = 1/2$ by continuation. Its sign agrees with $g''_a(0)$ for $a > 1/2$. The unscaled density is not defined at $a = 1/2$.

Step 4, the Hankel test. This step uses no Fourier analysis.

Lemma B.1 (a Hankel test)**.** *Let $s_j = (2j+1)\mathbb{E}\mathcal{L}_a^{2j}$. Suppose that for some $k \ge 1$ and $x \in \mathbb{R}^{k+1}$ we have $\sum_{i,j\le k} x_i x_j s_{i+j} < 0$ or $\sum_{i,j\le k} x_i x_j s_{i+j+1} < 0$. Then case (a) of Theorem 3.1 fails, so case (b) holds.*

Proof. In case (a), $\mathcal{L}_a^\circ$ is unimodal about 0. By Khinchine's theorem [15] it has the law of UZ, with U uniform on $(0,1)$ and independent of Z. So $s_j = (2j+1)\mathbb{E}(\mathcal{L}_a^\circ)^{2j} = \mathbb{E}W^j$ with $W = Z^2 \ge 0$, and the two sums are $\mathbb{E}(\sum_i x_i W^i)^2 \ge 0$ and $\mathbb{E}W(\sum_i x_i W^i)^2 \ge 0$. □

The script `certify_threshold.py` checks every hypothesis in ball arithmetic [14]. It evaluates R with a as a ball over I and evaluates G on subintervals of I, bisecting until $G \pm \eta$ has one sign on each. For the Hankel test it takes k and a rational x from a floating-point eigenvector at the center of a box of a, and checks the sum on the whole box with a Taylor model as in Step 2. Floating-point choices of N, the precision and x only affect whether the subsequent certificate succeeds. The cells have half-width $\frac{1}{32}$, the Taylor order in t is 12, $\beta = 4$, and the degree K is chosen so that $M\vartheta^{K+1} \le e^{-60}$. Table 3 lists the runs.

range of a	method	pieces	parameters	time
$[\frac{1}{2}, 0.668]$	Steps 1 to 3	40	$\vartheta = \frac{1}{8}$, $T_2 \in [3, 10]$	2.1 h
$[0.668, 0.6712]$	Steps 1 and 2	8	$\vartheta = \frac{1}{32}$, $\rho \le \frac{1}{100}$, $T_2 \in [4, 10]$	13 min
$[0.6705, 1]$	Step 4	115	$k \le 56$	25 min

Table 3: The runs of the certificate. The times are summed over the processes of each run.

Step 5, the unique transition. On $I = [0.66881285, 0.66881287]$, which contains $[z_-, z_+]$, the differentiated smoothing equation propagates bounds on $|\partial_t \phi_a|$ and $|\partial_a \phi_a|$. Section C.4 proves these bounds and the parameter Taylor estimates. The resulting certificate gives

$$0.68 < \partial_a g''_a(0) < 0.70, \qquad -0.09 < g_a^{(4)}(0) < -0.07 \quad (a \in \operatorname{int} I).$$

Proof of Theorem 3.4. The 40 intervals of the first run cover $[\frac{1}{2}, 0.668]$. On each, the scaled enclosure from Step 3 is negative, so $g_a''(0) < 0$ for $a \in (\frac{1}{2}, 0.668]$. The 8 intervals of the second run cover $[0.668, 0.6712]$. On the 2 intervals that cover $[0.668, 0.6688]$ the enclosure is negative, and on the 5 that cover $[0.6692, 0.6712]$ it is positive. On $[0.6688, 0.6692]$ we have $\eta < 1.1 \cdot 10^{-15}$. There the enclosure is negative on closed pieces that end at z_- and positive on closed pieces that start at z_+, and only (z_-, z_+) is undecided. The script prints these signs and checks this arrangement. So $g_a''(0) < 0$ on $(\frac{1}{2}, z_-]$ and $g_a''(0) > 0$ on $[z_+, 0.6712]$. The 115 boxes of the third run cover $[0.6705, 1]$, and on each the first sum of Lemma B.1 is negative.

By Step 1, g_a is even with two continuous derivatives, so $g_a'(0) = 0$. If $g_a''(0) < 0$, then $g_a' < 0$ on some $(0, \delta)$, which rules out case (b), where g_a is nondecreasing on $[0, \kappa]$. If $g_a''(0) > 0$, then $g_a' > 0$ on some $(0, \delta)$, which rules out case (a). Lemma B.1 gives case (b) on $[0.6705, 1)$. By Step 2, $a \mapsto g_a''(0)$ is continuous on $[0.6688, 0.6692]$, so it has a zero in (z_-, z_+). Step 5 makes it strictly increasing throughout that bracket, so the zero a_* is unique. The signs already proved exclude other zeros in $(\frac{1}{2}, 0.6712]$ and give the stated shape on either side of a_*. At a_*, evenness and Step 5 give $g_{a_*}'(x) = g_{a_*}^{(4)}(0)x^3/6 + o(x^3) < 0$ for small positive x. This excludes case (b), so the critical density is unimodal too. □

C Proofs for the threshold certificate

We prove the certificate bounds used in Appendix B, keeping its notation and the smoothing inequality (B.1). Together these bounds prove Theorem 3.4. We write $\Phi_N(t) = \sum_{n \le N} c_n (it)^n$ for the partial sums of the moment series.

C.1 Bounds at a fixed memory

Lemma C.1 (moment bounds)**.** *Let $N \ge 3$.*

(a) Put $A = ((N+1)a - 1)/N$. If $m_j \le AB^j$ for $1 \le j \le N$, then $m_n \le AB^n$ for all $n \ge 1$.

(b) Let $S \subset \mathbb{C}$ be compact, with $\frac{1}{2} < \alpha \le \operatorname{Re} z$, $|\operatorname{Im} z| \le \rho$ and $|z| \le 1$ for $z \in S$. Define $m_n(z)$ by the recursion with z in place of a, and $c_n(z) = m_n(z)/\Gamma(1 + nz)$. Then every m_n is analytic near S. Put $A = ((N+1)\alpha - 1)/N$. If $|m_j| \le AB^j$ on S for $1 \le j \le N$, then for all $n \ge 1$ and $z \in S$

$$|m_n(z)| \le AB^n, \qquad |c_n(z)| \le \frac{\tilde{A}\tilde{B}^n}{\Gamma(1 + n\alpha)}, \qquad \tilde{A} = Ae^{\rho^2/(2\alpha^2)}, \quad \tilde{B} = Be^{\rho^2/(2\alpha)}.$$

Proof. For (a) we use induction on $n > N$. As $\theta_j \le 1$ and $na - \theta_n \ge na - 1 > 0$, the bound for smaller indices gives $m_n \le (n-1)A^2B^n/(na-1)$. The ratio $(na-1)/(n-1)$ increases in n, since its derivative is $(1-a)/(n-1)^2$, and it equals A at $n = N+1$. So $m_n \le AB^n$.

For (b), $\operatorname{Re}(nz - \theta_n) \ge n\alpha - 1 > 0$ on S, because $\alpha \le 1$. So m_n is analytic near S, and the induction of (a) goes through with $|\theta_j| \le 1$. For $x > 0$ and real y,

$$\Big|\frac{\Gamma(x)}{\Gamma(x+iy)}\Big|^2 = \prod_{k \ge 0}\Big(1 + \frac{y^2}{(x+k)^2}\Big) \le \exp\Big(\frac{y^2}{x^2} + \frac{y^2}{x}\Big)$$

by [17, 5.8.3]. We take $x = 1 + n\operatorname{Re} z \ge 1 + n\alpha$ and $|y| \le n\rho$. Then $y^2/x^2 \le \rho^2/\alpha^2$ and $y^2/x \le n\rho^2/\alpha$. Finally $\Gamma(1 + n\operatorname{Re} z) \ge \Gamma(1 + n\alpha)$, as Γ increases on $[1.47, \infty)$. □

Lemma C.2 (truncation)**.** *Let $0 < \alpha \le 1$, and suppose that $|c_n| \le \tilde{A}\tilde{B}^n/\Gamma(1 + n\alpha)$ for $n > N$. Let $t > 0$, $0 \le k \le N$ and $n_0 = N + 1$, and put*

$$u = \frac{\tilde{A}\tilde{B}^{n_0} n_0^k t^{n_0 - k}}{\Gamma(1 + \alpha n_0)}, \qquad r = \tilde{B}t\Big(1 + \frac{1}{n_0}\Big)^k \frac{\Gamma(1 + \alpha n_0)}{\Gamma(1 + \alpha n_0 + \alpha)}.$$

If $r<1$, then $\left|\frac{d^k}{dt^k}\sum_{n>N} c_n(it)^n\right| \le \tau_k(t) := u/(1-r)$, and τ_k increases on $(0,t]$.

Proof. The kth derivative of $(it)^n$ has modulus at most $n^k t^{n-k}$. The ratio of consecutive terms of the majorant $\tilde{A}\tilde{B}^n n^k t^{n-k}/\Gamma(1+\alpha n)$ is $\tilde{B}t(1+1/n)^k\Gamma(1+\alpha n)/\Gamma(1+\alpha n+\alpha)$. It decreases in n, because $\log\Gamma$ is convex. So the majorant is at most a geometric series with first term u and ratio r. Both u and r increase in t. □

Lemma C.3 (frequency cells)**.** *Let ν be even, $t_c>0$ and $|\delta|\le h$. Then*

$$|\phi(t_c+\delta)| \le \sum_{k<\nu} |\phi^{(k)}(t_c)|h^k/k! + c_\nu h^\nu.$$

Proof. This is Taylor's formula, with $|\phi^{(\nu)}| \le \mathbb{E}|\mathcal{L}_a|^\nu = \mathbb{E}\mathcal{L}_a^\nu = \nu!\, c_\nu$. □

Lemma C.4 (Fourier tail)**.** *Let $0<T_2<T_3$ with $T_3 \ge 2^a T_2$, and let $\beta>3$ with $a\beta>1$. Cover $[0,T_3]$ by cells $[l_j,r_j]$ with $l_0=0$ and with T_2 among the end points, and let $s_j \ge \sup_{[l_j,r_j]}|\phi|$. Let C be the largest $s_j r_j^\beta$ over the cells that meet $[T_2,T_3]$, and put*

$$R = T_2^{-1/a}\sum_{r_j\le T_2} s_j\Big(1-\frac{1}{2}\Big(\frac{r_j}{T_2}\Big)^{1/a}\Big)^{-a\beta}\big(r_j^{1/a}-l_j^{1/a}\big) + \frac{2^{a\beta}CT_2^{-\beta}}{a\beta-1}.$$

If $R<1$, then $|\phi(t)| \le Ct^{-\beta}$ for every $t\ge T_3$.

Proof. Let $b=s_j$ on the cells in $[0,T_3)$ and $b(t)=Ct^{-\beta}$ for $t\ge T_3$. On a cell that meets $[T_2,T_3]$ we have $s_j \le Cr_j^{-\beta} \le Ct^{-\beta}$. So $b(t)\le Ct^{-\beta}$ for $t\ge T_2$, and $|\phi|\le b$ on $[0,T_3]$. Let $J(t)=\int_0^1 b(tv^a)\,b(t(1-v)^a)\,dv$.

We show that $J(t)\le Rb(t)$ for $t\ge T_3$. By symmetry $J(t)$ is twice the integral over $v\le\frac12$. There $t(1-v)^a \ge 2^{-a}T_3 \ge T_2$, so $b(t(1-v)^a)\le b(t)(1-v)^{-a\beta}$. Hence $J(t)/b(t)$ is at most $2\int_0^{1/2} b(tv^a)(1-v)^{-a\beta}\,dv$. Put $v_0=(T_2/t)^{1/a}\le\frac12$. On $v<v_0$ we substitute $u=tv^a$, and this part becomes

$$\frac{2}{a}t^{-1/a}\int_0^{T_2} s(u)\big(1-(u/t)^{1/a}\big)^{-a\beta}u^{1/a-1}\,du.$$

Here $t^{-1/a}\le\frac12 T_2^{-1/a}$ and $(u/t)^{1/a}\le\frac12(u/T_2)^{1/a}$. The weight increases in u, so on each cell we take it at r_j. As $\int_l^r u^{1/a-1}du = a(r^{1/a}-l^{1/a})$, this part is at most the first term of R. On $v\ge v_0$ we use $b(tv^a)\le Ct^{-\beta}v^{-a\beta}$ and $(1-v)^{-a\beta}\le 2^{a\beta}$. This part is at most $2^{1+a\beta}T_2^{-\beta}(T_2/t)^{1/a}C/(a\beta-1)$, which is at most the second term of R.

Let ς be the supremum of the $\tau\ge T_3$ with $|\phi|\le b$ on $[0,\tau]$, and suppose that $\varsigma<\infty$. By continuity $|\phi|\le b$ on $[0,\varsigma]$. The arguments in (B.1) at $t=\varsigma$ lie in $[0,\varsigma]$, so $|\phi(\varsigma)|\le J(\varsigma)\le Rb(\varsigma)<b(\varsigma)$. As $|\phi|$ and b are continuous on $[T_3,\infty)$, we get $|\phi|<b$ near ς, against the choice of ς. □

This is an explicit form of Step 4 in the proof of [9, Theorem 2.5], with computed constants.

Lemma C.5 (the second derivative)**.** *If $|\phi(t)|\le Ct^{-\beta}$ for $t\ge T_3$, with $\beta>3$, then g_a has two continuous derivatives and*

$$g_a''(0) = -\frac{1}{\pi}\Big(\sum_{\substack{n\le N\\ n\ \text{even}}}(-1)^{n/2}c_n\frac{T_3^{n+3}}{n+3}+E\Big), \qquad |E| \le \frac{T_3^3}{3}\tau_0(T_3) + \frac{CT_3^{3-\beta}}{\beta-3}.$$

Proof. As $t^2\operatorname{Re}\phi(t)$ is integrable, Fourier inversion gives $g_a(x)=\frac{1}{2\pi}\int e^{-itx}\operatorname{Re}\phi(t)\,dt$, and dominated convergence allows two derivatives under the integral [16, Section 3.3]. So $g_a''(0) = -\frac{1}{\pi}\int_0^\infty t^2\operatorname{Re}\phi(t)\,dt$. On $[0,T_3]$ we replace ϕ by Φ_N, whose real part is $\sum_{n\text{ even}}(-1)^{n/2}c_n t^n$, and integrate term by term. By Lemma C.2 the error is at most $\int_0^{T_3} t^2\tau_0(t)\,dt \le \frac13 T_3^3\tau_0(T_3)$. The integral over $t>T_3$ is at most $CT_3^{3-\beta}/(\beta-3)$. □

C.2 Bounds on memory intervals

Direct interval evaluation loses too much to cancellation. Near the threshold the series for ϕ cancels terms of size about 10^{60}, requiring intervals in a narrower than about 10^{-65}. We retain this cancellation by expanding in a around one exact point.

Lemma C.6 (Cauchy remainder)**.** *Let h be analytic near the closed disc $|z-a_0|\le\rho$, with $|h|\le M$ there and Taylor coefficients h_i at a_0. If $0<\vartheta<1$ and $|a-a_0|\le\vartheta\rho$, then $|h(a)-\sum_{i\le K}h_i(a-a_0)^i|\le M\vartheta^{K+1}/(1-\vartheta)$.*

Proof. Cauchy's estimate gives $|h_i|\le M\rho^{-i}$. Sum the geometric tail. □

Let $I=[a_0-w,a_0+w]$, $0<\vartheta<1$ and $\rho=w/\vartheta$, and let S be the square $a_0+[-\rho,\rho]+i[-\rho,\rho]$, which contains the disc of Lemma C.6. Suppose that $\alpha=a_0-\rho>\frac12$, that $|z|\le1$ on S, and that $\tilde A,\tilde B$ are as in Lemma C.1(b). Let τ_k be the bound of Lemma C.2 with these constants. It holds for every real $a\in I$. For a cell with center t_c write $\partial_t^k\Phi_N(z,t_c)=E_k(z)+iO_k(z)$, where E_k is the sum over even n and iO_k the sum over odd n. Both E_k and O_k are real for real z. Let $E_{k,i}$ and $O_{k,i}$ be their Taylor coefficients at a_0, and put $M_k=\sum_{n\le N}\sup_S|c_n|\,n(n-1)\cdots(n-k+1)\,t_c^{n-k}$.

Proposition C.7 (enclosure on an interval)**.** *(a) For every $a\in I$ and every t in the cell with center t_c and half-width h,*

$$|\phi_a(t)|\le\sum_{k<12}\Big(\sum_{i\le K}w^i\sqrt{E_{k,i}^2+O_{k,i}^2}+\frac{M_k\vartheta^{K+1}}{1-\vartheta}+\tau_k(t_c)\Big)\frac{h^k}{k!}+h^{12}\sup_S|c_{12}|.$$

(b) Let T_2, $T_3\ge2^{a_0+w}T_2$, β and the cells be as in Lemma C.4, with s_j the bound of (a), and suppose that $R<1$ for every $a\in I$. Put

$$F(z)=\sum_{n\le N,\,n\ \text{even}}(-1)^{n/2}c_n(z)T_3^{n+3}/(n+3).$$

Let F_i be its Taylor coefficients at a_0, and let M_F be the same sum with $\sup_S|c_n|$ in place of $(-1)^{n/2}c_n(z)$. Then for every $a\in I$

$$\Big|g_a''(0)+\frac1\pi\sum_{i\le K}F_i(a-a_0)^i\Big|\le\eta=\frac1\pi\Big(\frac{M_F\vartheta^{K+1}}{1-\vartheta}+\frac{T_3^3}{3}\tau_0(T_3)+\frac{CT_3^{3-\beta}}{\beta-3}\Big),$$

and $a\mapsto g_a''(0)$ is continuous on I.

Proof. For (a) we apply Lemma C.3 with $\nu=12$, and Lemma C.2 to replace $\phi_a^{(k)}$ by $\partial_t^k\Phi_N$. For real a we have $|\partial_t^k\Phi_N(a,t_c)|=|(E_k(a),O_k(a))|$. Lemma C.6 for E_k and O_k, with $\sup|E_k|+\sup|O_k|\le M_k$ on S, and the triangle inequality in $\mathbb{R}^2$ give the bracket. Also $c_{12}(a)\le\sup_S|c_{12}|$.

For (b), $T_3\ge2^aT_2$ for every $a\in I$, so Lemma C.4 applies at each a. Then Lemma C.5 and Lemma C.6 for F give the enclosure. By Lemma C.1(b) the series for $\phi_a(t)$ converges uniformly for $a\in I$ and t in compact sets. So $\phi_a(t)$ is continuous in a. Moreover $t^2|\phi_a(t)|$ is at most t^2 on $[0,T_3]$ and $Ct^{2-\beta}$ beyond, uniformly in $a\in I$. Dominated convergence gives the continuity. □

Near $a=\frac12$ the bounds of Lemma C.1 degenerate, since $m_2=a/(2a-1)$. We rescale.

Lemma C.8 (the scaled variable)**.** *Put $y=2a-1$ and $\mu_n=m_ny^{\lfloor n/2\rfloor}$.*

(a) We have $\mu_0 = \mu_1 = 1$, $\mu_2 = a$, *and for* $n \geq 3$

$$(na - \theta_n)\mu_n = \sum_{j=1}^{n-1} \theta_j y^{e_{nj}} \mu_j \mu_{n-j},$$

where $e_{nj} = 1$ *if* n *is even and* j *is odd, and* $e_{nj} = 0$ *otherwise. Here* $na - \theta_n$ *is* $(n-1)a$ *for odd* n *and* $na - 1$ *for even* n. *So every* μ_n *is a rational function of* a *with no pole in* $\operatorname{Re} a > \frac{1}{4}$.

(b) Let $\sigma^2 = \mathbb{E}\mathcal{L}_a^2$, $X = \mathcal{L}_a/\sigma$ *and* $\hat{c}_n = \mu_n \Gamma(2a)^{\lfloor n/2 \rfloor}/\Gamma(1+an)$. *Then* $\sigma^2 = 1/(y\Gamma(2a))$,

$$\mathbb{E}e^{iuX} = \sum_{n \text{ even}} \hat{c}_n (iu)^n + \sqrt{y\Gamma(2a)} \sum_{n \text{ odd}} \hat{c}_n (iu)^n, \qquad g_a''(0) = -\frac{1}{\pi\sigma^3} \int_0^\infty u^2 \operatorname{Re} \mathbb{E}e^{iuX}\, du.$$

Proof. Multiply the recursion for m_n by $y^{\lfloor n/2 \rfloor}$. The exponent $\lfloor n/2 \rfloor - \lfloor j/2 \rfloor - \lfloor (n-j)/2 \rfloor$ is 1 when n is even and j is odd, and 0 otherwise. Also $m_2 = a/y$. This proves (a), as the zeros of the factors are $a = 0$ and $a = 1/n \leq \frac{1}{4}$. For (b), $\sigma^2 = 2m_2/\Gamma(1+2a) = 1/(y\Gamma(2a))$ and $\mathbb{E}X^n/n! = c_n \sigma^{-n} = \mu_n \Gamma(2a)^{\lfloor n/2 \rfloor} (y\Gamma(2a))^{(n \bmod 2)/2}/\Gamma(1+an)$. The formula for $g_a''(0)$ follows from the substitution $t = u/\sigma$ in Lemma C.5. □

The fixed-point equation is linear, so X satisfies it too, and (B.1) and Lemmas C.3 to C.5 hold for X. The proof of Lemma C.1(b) carries over to μ_n for $n \geq 3$ and $\alpha > \frac{1}{4}$, with two changes. The constant A becomes $((N+1)\alpha - 1)/(N\kappa)$, where κ is the largest value of $\max(1, |z|)\max(1, |2z-1|)$ on S. And $\tilde{B}$ gains the factor $\max(1, \sup_S |\Gamma(2z)|)^{1/2}$. For real a the odd part carries the factor $\sqrt{y\Gamma(2a)}$. In the cell bounds we treat the even and odd parts as separate analytic functions. We multiply the odd part by $f = \sup_I \sqrt{y\Gamma(2a)}$ only for real a, and the remainders by $\max(1, f)$. The real part of $\mathbb{E}e^{iuX}$ involves only the even part. So the disc may cross $a = \frac{1}{2}$, and the cover reaches $a = \frac{1}{2}$. The sign of $g_a''(0)$ is the opposite of the sign of the integral, since $\sigma^3 > 0$ and the Fourier formula has a minus sign.

C.3 Checking the parameter cover

The script accepts an interval of a only when $G \pm \eta$ has one sign on each piece of a subdivision of it. In the 7 scaled runs every piece of every interval is negative, so $g_a''(0) < 0$ on $(\frac{1}{2}, 0.668]$. The zero run has 8 intervals that cover $[0.668, 0.6712]$. On the 2 intervals that cover $[0.668, 0.6688]$ every piece is negative. On the 5 intervals that cover $[0.6692, 0.6712]$ every piece is positive. On $[0.6688, 0.6692]$ the pieces up to z_- are negative, the pieces from z_+ on are positive, and only (z_-, z_+) is undecided. Proposition C.7(b) turns these signs into $g_a''(0) < 0$ on $(\frac{1}{2}, z_-]$ and $g_a''(0) > 0$ on $[z_+, 0.6712]$. To check the sign classification in Theorem 3.4, the script records each accepted interval with its sign and exact endpoints. Its mode `zero` fails unless the signs are $-$ before the undecided piece and $+$ after it. Its mode `check` requires the same of the stored records, together with the sign $-$ on every interval of the scaled runs.

C.4 Uniqueness of the transition

To prove uniqueness inside the bracket, we differentiate the smoothing equation and bound the derivative's tail. Differentiating the error bound in Lemma C.5 would not bound the derivative of the error.

Lemma C.9 (differentiated smoothing bound)**.** *Fix a compact memory interval* $I \subset (1/2, 1)$. *Suppose that the cell certificate of Lemma C.4 gives, uniformly on* I, $f(t) :=$

$|\phi_a(t)| \le Ct^{-\beta}$ *for* $t \ge T_2$, *and*

$$H_\beta(t) := 2\int_0^{1/2} f(tv^a)(1-v)^{-a\beta}\,dv \le R < 1 \quad (t \ge T), \qquad T \ge 2^a T_2.$$

Let $\beta > 4$, $\gamma = \beta - 1$, $a\gamma > 1$, *and* $u = |\partial_t \phi_a|$, $w = |\partial_a \phi_a|$. *Suppose that* $u \le M$ *everywhere,* $w \le M_a$ *on* $[0, T_2]$, *and cells give* $u(t) < D_1 t^{-\beta}$ *and* $w(t) < D_a t^{-\gamma}$ *on* $[T_2, T]$. *Put*

$$A_1 = \frac{MC2^{a(\beta-1)}}{a+1}, \qquad J = 2C2^{a\beta}T^{\gamma-\beta-1/a},$$
$$\kappa = R + \frac{JT_2^{1/a-\gamma}}{a\gamma - 1}, \qquad A_0 = JM_aT_2^{1/a},$$
$$A_S = MC2^{a(\beta-1)}\left\{\frac{\log 2}{a+1} + \frac{1}{(a+1)^2}\right\} + D_1(\log 2)R + CR/T.$$

If $A_1 + RD_1 < D_1$ *and* $A_0 + A_S + \kappa D_a < D_a$, *with* $\kappa < 1$, *then the two derivative bounds hold for every* $t \ge T_2$. *Consequently, in the interior of* I,

$$\frac{d}{da}g_a''(0) = -\frac{1}{\pi}\int_0^\infty t^2 \operatorname{Re}\partial_a\phi_a(t)\,dt, \qquad \left|\frac{1}{\pi}\int_T^\infty t^2 \operatorname{Re}\partial_a\phi_a(t)\,dt\right| \le \frac{D_aT^{3-\gamma}}{\pi(\gamma-3)}. \tag{C.1}$$

Proof. The complex majorant in Lemma C.1(b) makes the moment series locally uniformly convergent in the frequency and memory. Cauchy's estimate on a smaller parameter disc gives the same conclusion for its memory derivative and every fixed frequency derivative. Thus these are actual derivatives of ϕ_a. The smoothing equation, with $p = (1+a)/2$, is

$$\phi_a(t) = \int_0^1 \phi_a(tv^a)\{p\phi_a(t(1-v)^a) + (1-p)\phi_a(-t(1-v)^a)\}\,dv.$$

On compact frequency sets, its differentiation under the integral is justified by local boundedness of these derivatives and boundedness of $v^{\inf I}|\log v|$. Conjugate symmetry gives even moduli. Writing $x = tv^a$ and $y = t(1-v)^a$, we obtain

$$u(t) \le \int_0^1 \{v^a u(x)f(y) + (1-v)^a f(x)u(y)\}\,dv,$$
$$w(t) \le \int_0^1 \{w(x)f(y) + f(x)w(y)\}\,dv + S(t),$$
$$S(t) \le \int_0^1 \{x|\log v|u(x)f(y) + y|\log(1-v)|f(x)u(y) + f(x)f(y)\}\,dv.$$

The final term includes the derivative of p. Its coefficient is one because $p' = 1/2$ and $|\phi_a(y) - \phi_a(-y)| \le 2f(y)$.

Split each integral at $1/2$ and reflect its second half. For the frequency derivative, the smaller-argument term is at most $A_1t^{-\beta}$, by $u \le M$ and

$$2\int_0^{1/2} v^a(1-v)^{-a\beta}\,dv \le \frac{2^{a(\beta-1)}}{a+1}.$$

The larger-argument term is at most $RD_1t^{-\beta}$. For the memory derivative, split the smaller-argument term again at $v_0 = (T_2/t)^{1/a}$. Its bound is

$$2C2^{a\beta}t^{-\beta-1/a}\left\{M_aT_2^{1/a} + \frac{D_aT_2^{1/a-\gamma}}{a\gamma-1}\right\}.$$

The larger-argument term is at most $RD_a t^{-\gamma}$, since $H_\gamma \le H_\beta$. As $\gamma - \beta - 1/a < 0$ and $t \ge T$, these two terms are bounded by $(A_0 + \kappa D_a)t^{-\gamma}$. In S, the smaller logarithmic term uses

$$\int_0^{1/2} v^a(-\log v)\,dv = 2^{-a-1}\left\{\frac{\log 2}{a+1} + \frac{1}{(a+1)^2}\right\};$$

the larger uses $-\log(1-v) \le \log 2$ and $H_\beta \le R$. The mixing-weight term is at most $CRt^{-\beta}$. Thus $S(t) \le A_S t^{-\gamma}$.

To extend the bounds beyond the cells, maximize the continuous weighted derivative on any $[T_2, U]$. If its maximum lies in $[T_2, T]$, use the cell bound. Otherwise repeat the preceding inequalities with that maximum in place of the proposed constant, and rearrange the strict contraction inequality. This first proves the frequency bound, then the memory bound, independently of U. The integrable uniform bound $t^2 w(t) \le D_a t^{2-\gamma}$ now permits differentiation of the Fourier formula for $g_a''(0)$ on the interior of I and gives (C.1). □

The computation uses

$$I = [0.66881285, 0.66881287], \quad T_2 = 6, \quad T = 10, \quad \beta = 8, \quad \gamma = 7,$$

and the conservative constants

$$C = 17/1000, \quad R = 83/1000, \quad M = 2, \quad D_1 = 2, \quad M_a = D_a = 4.$$

It checks $A_1 < 1$, $\kappa < 1/10$ and $A_0 + A_S < 2$, so both contractions have strict slack. The new tail is at most $1/(10000\pi) < 1/30000$. The certified zero bracket lies strictly inside I.

For the finite-frequency integral, let w_0 be the halfwidth of I, $\rho > w_0$ the Cauchy radius and $\vartheta = w_0/\rho$. Differentiating the finite Taylor polynomial in memory leaves the coefficient remainder factor

$$\frac{1}{\rho}\sum_{i>K} i\vartheta^{i-1} = \frac{\vartheta^K\{(K+1) - K\vartheta\}}{\rho(1-\vartheta)^2}.$$

The omitted moment series has derivative bound $\tau_k(t)/(\rho - w_0)$ by Cauchy's estimate. On each frequency cell we use a Taylor expansion of order 48, with its remainder bounded by the complex coefficient majorant. (At complex memory we cannot use a probability moment.) The signed interval checks give

$$0.68 < \frac{d}{da} g_a''(0) < 0.70, \qquad -0.09 < g_a^{(4)}(0) < -0.07 \quad (a \in \operatorname{int} I). \tag{C.2}$$

The second enclosure uses the same Fourier bound with weight t^4, whose tail is $CT^{-3}/(3\pi)$. The file `certify_transition.py` performs these checks, including the initial derivative-cell bounds. It was rerun with 1100 moments and 1600-bit precision under `python -O`. Every check uses `require`, so optimization cannot remove it.

Continuity and the sign bracket give a zero a_* of $g_a''(0)$. The first inequality in (C.2) makes it unique within I. At a_*, evenness and the second inequality give

$$g_{a_*}'(x) = \tfrac{1}{6} g_{a_*}^{(4)}(0) x^3 + o(x^3) < 0 \quad \text{for all sufficiently small } x > 0.$$

The strict shape dichotomy therefore puts a_* in the unimodal branch. The signs inside I and the shape certificate outside the bracket now prove the unique transition in Theorem 3.4. Central curvature could have further zeros in the bimodal region without changing this conclusion.

The critical constants. For Table 1 we apply the same estimates on $I_* = [z_- - 10^{-14}, z_- + 10^{-14}]$, where $z_- = 1838418790980431/2748779069440000$ is the lower endpoint of the certified threshold bracket. The entire bracket lies inside I_*. Take

$$T_2 = 10, \quad T = 16, \quad \beta = 12, \quad \gamma = 11,$$
$$C = 1/1000, \quad R = 1/20, \quad M = D_1 = 2, \quad M_a = D_a = 4.$$

The base Fourier bound, initial derivative cells and both strict contraction inequalities hold uniformly on I_*. We use 2000 moments, 2600-bit ball arithmetic, Cauchy ratio $1/10000$, frequency halfwidth $1/32$ and frequency Taylor order 48. The omitted tails for C_*, B_* and H_* are bounded respectively by

$$\frac{D_a T^{3-\gamma}}{\pi(\gamma-3)}, \qquad \frac{CT^{5-\beta}}{\pi(\beta-5)}, \qquad \frac{CT^{2-\beta}}{\pi(\beta-2)}.$$

The finite-frequency integrals use the same parameter and moment remainders as above, with weights t^2, t^4 and t. Since the bounds hold on the whole interval, they include a_* even though its exact value is unknown. Outward rounding gives the table's two derived coefficients. The record and rerun command are in Appendix F.

D Signs of the second difference

We prove the sign bounds of Section 2.

D.1 Averaging adjacent entries

Insert each η_i between its two neighbors in ζ. If $\eta_i = 0$, the sign pattern does not change. If $\eta_i > 0$, then $\zeta_i > 0$ or $\zeta_{i+1} > 0$, so η_i is placed next to an entry of its own sign, and the pattern does not change. The same argument applies when $\eta_i < 0$, proving the first claim. Deleting entries from a list replaces its sign pattern by a subsequence, and η is obtained from w by deleting the entries ζ_i. This proves the second claim. □

D.2 Signs at the first lattice point

Put $\xi_t = P_t(1)/P_t(0)$. By (2.1), $P_{t+1}(0) = (1-\varepsilon)P_t(0)$ and $P_{t+1}(1) = (1-\varepsilon-\frac{a}{t})P_t(1) + \varepsilon P_t(0)$. So

$$\xi_{t+1} - \xi_t = \frac{1}{1-\varepsilon}\Big(\varepsilon - \frac{a\xi_t}{t}\Big), \qquad \xi_2 = \frac{2\varepsilon}{1-\varepsilon}.$$

We claim that $\xi_t < \varepsilon t/a$ for $t \geq 2$. For $t = 2$ this holds because $a < 1-\varepsilon$. If it holds for t, then $0 < \xi_{t+1} - \xi_t \leq \frac{\varepsilon}{1-\varepsilon} < \frac{\varepsilon}{a}$, and it holds for $t+1$. So ξ_t is strictly increasing for $t \geq 2$, and $\xi_t < 1$ whenever $t \leq a/\varepsilon$.

Now (2.2) with $k = 1$, and $D_t(0) = P_t(1) - 2P_t(0)$, give for $t \geq 2$

$$s_{t+1} = \big(1 - r_t(2)\big)s_t + \sigma_t, \qquad \sigma_t = \Big(\varepsilon - \frac{a}{t}\Big)(\xi_t - 2)\, P_t(0).$$

Here $1 - r_t(2) \geq 1 - \varepsilon - a = \varepsilon > 0$. If $t < a/\varepsilon$, both factors of σ_t are negative and $\sigma_t > 0$. If $t > a/\varepsilon$, the first factor is positive, and the second is first negative and then positive, because ξ_t increases. So $\mathrm{sp}(\sigma_2, \sigma_3, \dots) \preceq \Pi_3$. Let $\Phi_t = \prod_{u=2}^{t-1}(1 - r_u(2)) > 0$. Then

$$\frac{s_t}{\Phi_t} = s_2 + \sum_{u=2}^{t-1} \frac{\sigma_u}{\Phi_{u+1}}, \qquad s_2 = 1 - 3\varepsilon.$$

If $\varepsilon < \frac{1}{3}$, the terms $s_2, \sigma_2, \sigma_3, \dots$ are first nonnegative, then nonpositive, then nonnegative. Their partial sums are nonnegative and nondecreasing in the first stretch, nonincreasing

in the second, and nondecreasing in the third. So the sign pattern of the partial sums is a subsequence of Π_3. If $\varepsilon \geq \frac{1}{3}$, then $a/\varepsilon \leq 1$, so the terms are first nonpositive and then nonnegative, and the sign pattern of the partial sums is a subsequence of $(-,+)$. This proves the first claim.

For the second claim let $0 \leq \varepsilon \leq 1$. The proofs of (2.1) and (2.2) use only that step $t+1$ is $+1$ with probability $r_t(k)$ given $A_t = k$, so both hold for this ε. We compute $P_3(0) = \frac{1}{2}(1-\varepsilon)^2$ and $P_3(1) = P_3(2) = \frac{1}{2} - P_3(0)$, so $s_3 = P_3(0) - P_3(1) = (1-\varepsilon)^2 - \frac{1}{2}$. Suppose that $s_3 = 0$. Then $\varepsilon = 1 - 2^{-1/2}$ and $P_3(0) = P_3(1) = \frac{1}{4}$. By (2.2) with $t = 3$ and $k = 1$, $s_4 = r_3(-1)D_3(0) = (\varepsilon - \frac{a}{3})(P_3(1) - 2P_3(0)) = -\frac{5\varepsilon - 1}{12}$. This is not 0, because $\varepsilon \neq \frac{1}{5}$. (With P_t defined by (2.1), the same computation works for every real ε, because the other root $1 + 2^{-1/2}$ of s_3 is not $\frac{1}{5}$ either. We need only $0 < \varepsilon < \frac{1}{2}$.) □

D.3 At most five sign blocks

For $3 \leq t \leq N$ let $\mathcal{Z}_t$ be the list

$$\mathcal{Z}_t = \big(s_N, s_{N-1}, \ldots, s_{t+1},\ s_t, D_t(2), D_t(3), \ldots, D_t(t-2), s_t,\ s_{t+1}, \ldots, s_{N-1}, s_N\big).$$

It has $2N-t-1$ entries. Its middle part is $(D_t(1), D_t(2), \ldots, D_t(t-1))$, because $D_t(t-1) = D_t(1) = s_t$ by symmetry. The list $\mathcal{Z}_3$ is $(s_N, \ldots, s_4, s_3, s_3, s_4, \ldots, s_N)$, and $\mathcal{Z}_N$ is the list in the statement.

We check that $\mathcal{Z}_{t+1}$ is obtained from $\mathcal{Z}_t$ as in Lemma 2.2, for $3 \leq t \leq N-1$. The first $N-t$ entries of $\mathcal{Z}_{t+1}$ are $s_N, \ldots, s_{t+1}$, and they are copies of the first $N-t$ entries of $\mathcal{Z}_t$. So $A_i = 1$ and $B_i = 0$ there. The next $t-2$ entries are $D_{t+1}(b)$ for $2 \leq b \leq t-1$. By (2.2),

$$D_{t+1}(b) = r_t(b-2)\, D_t(b-1) + \big(1 - r_t(b+1)\big) D_t(b),$$

and $D_t(b-1)$ and $D_t(b)$ are neighbors in the middle part of $\mathcal{Z}_t$, at the right places. The coefficients satisfy $r_t(b-2) \geq \varepsilon$ and $1 - r_t(b+1) \geq 1 - \varepsilon - a = \varepsilon$, since $2 \leq b \leq t-1$. The last $N-t$ entries of $\mathcal{Z}_{t+1}$ are $s_{t+1}, \ldots, s_N$, and each is a copy of the entry of $\mathcal{Z}_t$ one place to its right. So $A_i = 0$ and $B_i = 1$ there.

By Lemma 2.2, $\mathrm{sp}(\mathcal{Z}_N) \preceq \mathrm{sp}(\mathcal{Z}_{N-1}) \preceq \cdots \preceq \mathrm{sp}(\mathcal{Z}_3)$. By Lemma 2.4 the sign pattern p of $(s_3, \ldots, s_N)$ is a subsequence of Π_3, so it is one of the lists $()$, $(+)$, $(-)$, $(+,-)$, $(-,+)$ and $(+,-,+)$. The sign pattern of $\mathcal{Z}_3$ is p reversed, followed by p, with equal neighbors merged. In the six cases this is $()$, $(+)$, $(-)$, $(-,+,-)$, $(+,-,+)$ and $(+,-,+,-,+)$, and each is a subsequence of Π_5.

□

E Derivative limits for a fixed start

We need first-difference convergence for the strict slope at zero. The same argument works at every fixed order and for any prescribed finite initial history. It uses the higher-difference recurrence of [10, (4.10)].

Lemma E.1. *Fix $a \in (1/2, 1)$ and a finite initial history. Suppose the rescaled count law converges weakly to a law with smooth density f. For each fixed integer $r \geq 0$, linearly interpolate*

$$\frac{n^{(r+1)a}}{2^{r+1}} \nabla^r P_n(k) \quad \text{at} \quad (2k - n - r)n^{-a}, \qquad \nabla P_n(k) = P_n(k) - P_n(k-1).$$

These interpolants converge locally uniformly to $f^{(r)}$.

Proof. Repeated subtraction in (2.1) gives

$$\nabla^r P_{n+1}(k) = \left(\frac{1+a}{2} - \frac{ak}{n}\right) \nabla^r P_n(k) + \left(\frac{1-a}{2} + \frac{a(k-r-1)}{n}\right) \nabla^r P_n(k-1).$$

For sufficiently large n, all coefficients multiplying nonzero entries are nonnegative. Interior row sums are $1-(r+1)a/n$. At either outer endpoint the only nonzero term has coefficient $(1+a)/2$, which is no larger once n is large. Iteration gives

$$\sup_k |\nabla^r P_n(k)| \le C_{a,r} n^{-(r+1)a}.$$

The bounds at orders r and $r+1$ give uniform bounds and Lipschitz constants for the interpolants. They are therefore compact for locally uniform convergence. Discrete summation by parts against a smooth compactly supported function identifies every subsequential limit with the distributional derivative of order r of the weak limit. The lattice spacing is $2n^{-a}$, so the displayed normalization is exactly its power $-(r+1)$. Since f is smooth, this distributional derivative is $f^{(r)}$. Every subsequence has the same limit, proving the assertion. □

F Data and certificate records

The tagged repository and submission archive contain the certificate code, records, source hashes and rerun commands. The parameter-cover records are data/threshold_*.txt, checked by certify_threshold.py. The transition derivative records use 1100 moments at 1600 bits, Python 3.12.14 and python-flint 0.8.0.

With python-flint 0.9.0, check_critical_shape.py checks the real threshold endpoints using 1800 moments at 2000 bits and the conditional slope using 1200 moments. certify_critical_constants.py encloses the critical constants using 2000 moments at 2600 bits. Both checks retain their acceptance tests under `python -O`. The numerical phase illustration and finite-length examples are not certificates. The README lists each script and its output.

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